Cheap Talk with Costly Information Acquisition by the Receiver

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Abstract

This paper examines the welfare implications of the receiver's costly information acquisition in a cheap-talk model. I find that both a reduction in the cost of information acquisition and an increase in the informativeness of the information source can generate welfare gains, regardless of the welfare weights or whether receiver learning crowds out sender communication. Furthermore, a cost reduction improves welfare under broader conditions than an increase in informativeness.

1 Introduction

The cheap-talk model proposed by Crawford and Sobel (1982) is widely applied to analyze the decision-making process in which a perfectly informed but biased expert (sender) is the only information source available to the decision-maker (receiver). This model has subsequently been extended to involve cases where the receiver has access to multiple free information sources, which are either biased (Krishna and Morgan, 2001; Ambrus and Lu, 2014) or unbiased (Watson, 1996; Olszewski, 2004; De Barreda, 2010). The receiver's access to information sources in addition to the sender has been shown to either hamper (Lai, 2014; Ishida and Shimizu, 2016) or enhance (Dziuda and Salas, 2018; Ishida and Shimizu, 2019) communication between the two parties. In particular, Balbuzanov (2019) showed that the effect of the receiver's extra free information on communication may vary across situations (in his setting, it depends on the bias of the sender).

In reality, people do not always have free access to information sources. Pei (2015); Argenziano et al. (2016), respectively, investigated cases where the sender is not perfectly informed and, instead, gathers costly information before sending a message to the receiver. Similarly, the receiver may have to acquire extra information at a cost in various situations.

For instance, when shopping on Amazon, it is time-consuming for customers to refer to historical reviews to know better about the items. Similarly, investors usually spend time and energy collecting and analyzing data about various investment tools before making investment decisions. The precision of the acquired information often depends on the receiver's acquisition effort, which is usually positively related to the relevant information cost. In such situations, the positive effect of the extra information and the consequent enhancement of communication (if any) may not fully offset the negative effect of the receiver's information cost on welfare (which is a weighted average of the payoff of the receiver and that of the sender). Thus, even if the receiver's costly information enhances communication between the two parties, the relevant total welfare effect need not be positive.

I depart from the current body of literature on cheap-talk models by investigating the total welfare effects of costly information sources available to the receiver. In particular, I extend the model of Crawford and Sobel (1982) (referred to as the CS model henceforth) by assuming that the receiver could, upon receiving the message from the perfectly informed sender, choose whether to acquire costly information and, if yes, decide on the level of her acquisition effort, which is positively related to both her information cost and the precision of the acquired information. I evaluate the costly information source in two aspects: cost of information acquisition and informativeness. The latter determines the highest possible precision of the acquired information. Correspondingly, I discuss the welfare effects of two types of changes in the information source: a reduction in the cost of information acquisition and an increase in the informativeness of the information source. I also pay special attention to a particular case of the second type of change where the receiver obtains access to an informative information source, i.e., the informativeness is increased from zero to positive. I decompose the total welfare effect of each type of change into two parts: direct effect and indirect effect. The direct effect holds the sender's messaging strategy to be independent of the change and only includes the welfare consequences of the subsequent changes in the receiver's choices (effort level and action). The indirect effect accounts for the welfare consequences of the reoptimization over the new set of equilibria after the change. It involves the strategic interactions between the receiver's choices and the optimization over the set of equilibrium messaging strategies.

I find that the direct effect of each type of change is always non-negative, and it is positive if and only if, after the change, the information source is not uninformative and the receiver exerts greater acquisition effort upon receiving some message (and, for a cost reduction, the welfare weight on the receiver is positive when the receiver exerts full effort upon receiving every message even before the change). The inability to explicitly characterize an equilibrium welfare maximizer makes it difficult to discuss the indirect effect in general,

but, for each type of change, the indirect effect is non-negative if the equilibrium welfare maximizer before the change is still supported by an equilibrium after the change. For each type of change, since the direct effect is always non-negative, the total effect will be non-negative if the indirect effect is also non-negative. Similarly, the total effect will be positive if the direct effect is positive and the indirect effect is non-negative, while the sign of the total effect will be unclear if the indirect effect is negative. Thus, the total effect of the receiver's access to an informative costly information source is always non-negative because any equilibrium messaging strategy in the CS model (the benchmark case, where no informative costly information source is available to the receiver) is still supported by an equilibrium when the receiver has access to the information source. Also, for both a cost reduction and an increase in informativeness, when the cost of information acquisition is not too low and the information source is not too informative, the indirect effect would be non-negative, which implies a non-negative total effect.

This paper contributes to the relatively small strand of literature on cheap-talk models with the receiver's costly information acquisition. Miyahara and Sadakane (2017) investigated a finite-state cheap-talk model where the receiver acquires three costly signals, one for each possible realization of the state. They showed that the receiver's information acquisition enhances communication. Sadakane and Tam (2026) also proposed an extension of cheap-talk models by involving a perfect lie-detection technology. They found that the receiver's access to the lie-detection technology not only enhances communication between the two parties but also increases the receiver's payoff in her optimal equilibrium. In contrast to these authors, I investigate an infinite-state cheap-talk model and focus on the welfare effect of the extra information acquired by the receiver from a costly information source that is, in general, imperfect. I discuss the welfare effects of multiple types of changes in the information source, including that of the receiver's access to an informative costly information source. My results are strong in that they hold for arbitrary welfare weights on the two parties, including the special case that only involves the receiver's payoff. The novelty of my paper as compared to the existing literature (Dziuda and Salas, 2018; Ishida and Shimizu, 2019) is that, even when the receiver only has access to a costly information source, the positive effect of the extra information is shown to, under some conditions, outweigh the negative effect of the information cost and induce welfare improvements regardless of the relevant change in the informativeness of communication between the two parties.

The remainder of this paper is organized as follows. Section 2 presents the model. Section 3 describes the solution concept adopted in the model and solves the model accordingly. Section 4 analyzes the welfare effects of different types of changes in the costly information source. Section 5 concludes. Proofs are all relegated to Appendix A.

2 Model

I consider a Bayesian game with two players: one uninformed receiver (she) and one perfectly informed sender (he). Their payoffs both depend on the state of nature (i.e., the sender's type) $\theta \in \Theta = [0,1]$ and the receiver's action $y \in \mathbb{R}$, while the receiver's payoff also depends on the level of her effort on information acquisition $q \in [0,1]$. To be more specific, the sender's payoff function is $u(y,\theta,b) := -(y-\theta-b)^2$, where $b > 0$ represents the bias in the sender's preference, and the receiver's payoff function is $v(y,\theta,q) := -(y-\theta)^2 - cq^2$, where cq^2 is the receiver's information cost with effort level q and parameter $c > 0$. Players hold common prior $\theta \sim U[0,1]$. The game proceeds as follows.

1. The sender observes his type θ, which is randomly drawn from $U[0,1]$, and then sends a message $m \in M = [0,1]$ to the receiver at no cost.

2. Upon receiving m, the receiver chooses an effort level $q \in [0,\bar{q}] \subset [0,1]$ and then receives a costly signal $s \in \Theta$ that is randomly drawn from a mixed distribution $(q \circ \delta_\theta, (1-q) \circ U[0,1])$.

3. Upon observing m and s, the receiver chooses an action $y \in \mathbb{R}$. Then, the game is over with payoffs realized according to the payoff functions.

The sender's strategy is a function $\sigma : \Theta \to \Delta(M)$ according to which he, in stage 1 of the game, chooses to randomize over the set of feasible messages M with distribution $\sigma(\theta)$ when his type is θ. I assume for simplicity that M is the same as the type space $\Theta = [0,1]$. The gist of this assumption is that M is infinite. It guarantees that M can support any equilibrium regardless of the way in which different types of the sender are pooled together and separated (whereas a finite M can not). The receiver's strategy is a pair of functions $q : M \to [0,\bar{q}]$ and $y : M \times \Theta \to \mathbb{R}$, where q is the effort level, y is the action, and $\bar{q} \in [0,1]$ is the receiver's highest possible effort level as well as the highest possible precision of the subsequent signal s, i.e., the informativeness of the costly information source.

In stage 1, given a strategy profile (σ, q, y), type θ sender's expected payoff of sending message m is

$$U(m|\theta) := q(m)u[y(m,\theta),\theta,b] + [1-q(m)]\int_0^1 u[y(m,s),\theta,b]ds,$$

where the first term on the right-hand side of the expression reflects the fact that the signal s reports the true state θ with probability $q(m)$, and the second term reflects the fact that s is randomly drawn from the common prior $U[0,1]$ with probability $1-q(m)$. The probability

distribution of the signal s in my model follows Ishida and Shimizu (2019) but differs from theirs in that (1) the probability $q = q(m)$ is chosen by the receiver at her own cost rather than being exogenously endowed for free and (2) the receiver chooses q upon receiving the sender's message, thus being able to choose different levels of precision of the extra signal upon receiving different messages from the sender. Note that when $\bar{q} = 0$, i.e., when the costly information source is uninformative, the information structure of my model becomes the same as that in the CS model, where the sender's message is the only information available to the receiver.

In stage 2, given a strategy profile (σ, q, y) and a belief system $\mu : M \to \Delta(\Theta)$, the receiver updates her belief upon receiving the message m, and her expected payoff of choosing effort level $q \in [0,1]$ is

$$V(q|m) := \int_0^1 \Big\{ qv[y(m,\theta'),\theta',q] + (1-q)\int_0^1 v[y(m,s),\theta',q]ds \Big\}\mu(\theta'|m)d\theta'.$$

In stage 3, given a strategy profile (σ, q, y) and a belief system $\mu : M \to \Delta(\Theta)$, the receiver updates her belief upon observing both the message m and the signal s, and her expected payoff of choosing action $y \in \mathbb{R}$ is

$$V(y|m,s) := \begin{cases} \int_0^1 v[y,\theta',q(m)]\mu(\theta'|m)d\theta' & \text{if } \mu(s|m) = 0 \\ \frac{q(m)v[y,s,q(m)]+[1-q(m)]p_\mu(m)\int_0^1 v[y,\theta',q(m)]\mu(\theta'|m)d\theta'}{q(m)+[1-q(m)]p_\mu(m)} & \text{if } \mu(s|m) > 0 \end{cases},$$

where $p_\mu(m) := Prob[\mu(\theta|m) > 0]$.

In this stage, the receiver uses Bayesian updating based on the realization of both m and s. To be more specific, when $\mu(s|m) = 0$, i.e., when the realization of s contradicts $\mu(.|m)$, the belief associated with m, the receiver always believes that the signal s is a randomly drawn noise. Thus, she updates her information only according to $\mu(.|m)$. When $\mu(s|m) > 0$, i.e., when the realization of s is consistent with $\mu(.|m)$, from the receiver's perspective, either of the following two cases may happen: (1) s is drawn from the truth-telling degenerate distribution δ_θ (with prior probability $q(m)$); (2) s is both randomly drawn (with prior probability $1-q(m)$) and not contradicting the belief $\mu(.|m)$ (with prior probability $p_\mu(m)$). Thus, the posterior probability of s's being drawn from δ_θ is $\frac{q(m)}{q(m)+[1-q(m)]p_\mu(m)}$, and that of s's being a random noise drawn from $U[0,1]$ is $\frac{[1-q(m)]p_\mu(m)}{q(m)+[1-q(m)]p_\mu(m)}$. The receiver's expected payoff is calculated according to these posterior probabilities.

The underlying assumption in the Bayesian updating described above is that the receiver always considers her belief system μ to be no less trustworthy than the signal s. This assumption is not invulnerable to caveats in all cases but fits fine in an equilibrium analysis,

where μ is derived from the equilibrium strategy profile. As a result, in equilibrium, the receiver always considers μ to be no less trustworthy than s, which is randomly drawn with non-negative probability.

3 Equilibrium

3.1 Solution Concept

The analysis of this model restricts attention to *monotone partition equilibria*[1] (Ishida and Shimizu, 2019), a subset of perfect Bayesian equilibria.

Definition 1. *A perfect Bayesian equilibrium* $((\sigma^*, q^*, y^*); \mu^*)$ *consisting of strategy profile* (σ^*, q^*, y^*) *and belief system* μ^* *is a monotone partition equilibrium if there exists a partition of* Θ*, denoted as* $\{\Theta_i\}_{i \in I}$*, so that:*

1. *for any* $i \in I$*,* Θ_i *is a non-empty interval;*
2. *for any* $i \in I$ *and any* $\theta, \theta' \in \Theta_i$*,* $\sigma^*(\theta) = \sigma^*(\theta')$*,* $\mathrm{supp}(\sigma^*(\theta)) =: M_i \neq \varnothing$*;*
3. *for any* $i, j \in I$ *so that* $i \neq j$*,* $M_i \bigcap M_j = \varnothing$*.*

Also, in my model, the definition of perfect Bayesian equilibrium accounts for the receiver's choice of effort level q as follows.

Definition 2. *An assessment* $((\sigma^*, q^*, y^*); \mu^*)$ *consisting of strategy profile* (σ^*, q^*, y^*) *and belief system* μ^* *is a perfect Bayesian equilibrium if:*

1. *for any state* θ *and any message* $m^* \in \mathrm{supp}(\sigma^*(\theta))$*,*

$$m^* \in \underset{m \in M}{\mathrm{argmax}}\, U(m|\theta); \tag{SR_m}$$

2. *for any message* m*, effort level* $q^*(m)$ *solves*

$$\max_{q \in [0, \bar{q}]} V(q|m); \tag{SR_q}$$

[1] Though Crawford and Sobel (1982) showed that, in their setting, all perfect Bayesian equilibria are equivalent to monotone partition equilibria, Chen (2009) showed that there may exist non-monotone equilibria in a cheap-talk model where the receiver has access to some information that is correlated with the sender's information (type). I focus on this specific class of equilibria because the main interest of this paper is the effect of the receiver's costly information on welfare, not that on the set of equilibria.

3. for any message m and any signal s, action $y^*(m,s)$ solves

$$\max_{y\in\mathbb{R}} V(y|m,s); \tag{SR_y}$$

4. for any state θ and any message m,

$$\mu^*(\theta|m) = \frac{\sigma^*(m|\theta)}{\int_\Theta \sigma^*(m|\theta)}. \tag{BR}$$

3.2 Characterization of Equilibrium

I find an equilibrium using backward induction.

First, for an equilibrium $((\sigma^*,q^*,y^*);\mu^*)$ with associated partition $\{\Theta_i\}_{i\in I}$ and any $i\in I$, since Θ_i is a non-empty interval, its closure can be written as $\bar{\Theta}_i = [\underline{\theta}_i, \bar{\theta}_i]$, where $\underline{\theta}_i = \inf(\Theta_i)$, $\bar{\theta}_i = \sup(\Theta_i)$, and $0 \le \underline{\theta}_i < \bar{\theta}_i \le 1$. Let $\Delta_i := \bar{\theta}_i - \underline{\theta}_i$, then $\Delta_i \in (0,1]$ is the length of interval Θ_i.

1. Stage 3 (SR_y).

For an equilibrium $((\sigma^*,q^*,y^*);\mu^*)$ with associated partition $\{\Theta_i\}_{i\in I}$, by BR, for any $\theta\in\Theta$, any $i\in I$, and any $m\in M_i$, with $\Delta_i\in(0,1]$,

$$\mu^*(\theta|m) = \frac{\mathbb{1}[\theta\in\Theta_i]}{\Delta_i}. \tag{BR'}$$

Thus, for any $i\in I$, any $m_i\in M_i$, any $s\in\Theta$, and any $y\in\mathbb{R}$,

$$V(y|m_i,s) = -\frac{\mathbb{1}[s\in\Theta_i]q^*(m_i)(y-s)^2 + [1-q^*(m_i)]\int_{\underline{\theta}_i}^{\bar{\theta}_i}(y-\theta)^2 d\theta}{\mathbb{1}[s\in\Theta_i]q^*(m_i) + [1-q^*(m_i)]\Delta_i} - cq^*(m_i)^2.$$

Accordingly, let $\hat{\theta}_i := \frac{\underline{\theta}_i+\bar{\theta}_i}{2}$, then, by SR_y, the receiver's optimal action is

$$y_i(s) := y^*(m_i,s) = \frac{\mathbb{1}[s\in\Theta_i]q^*(m_i)s + [1-q^*(m_i)]\Delta_i\hat{\theta}_i}{\mathbb{1}[s\in\Theta_i]q^*(m_i) + [1-q^*(m_i)]\Delta_i}. \tag{SR_y'}$$

2. Stage 2 (SR_q).

For an equilibrium $((\sigma^*,q^*,y^*);\mu^*)$ with associated partition $\{\Theta_i\}_{i\in I}$, by BR' and SR_y', for any $i\in I$, any $m_i\in M_i$, and any $q\in[0,1]$, with $\Delta_i\in(0,1]$,

$$V(q|m_i) = -cq^2 - \frac{(1-q)(q+\Delta_i)\Delta_i^2}{12[\Delta_i+(1-\Delta_i)q]}.$$

Thus, by SR_q, the receiver's optimal effort level is

$$q^*(m_i) = \begin{cases} 0 & \text{if } c \geq \frac{\Delta_i}{12} \\ \frac{\Delta_i(-48c+\Delta_i+\sqrt{96c\Delta_i+\Delta_i^2})}{48c(1-\Delta_i)} & \text{if } c \in [\frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}, \frac{\Delta_i}{12}) \\ \bar{q} & \text{if } c \in (0, \frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}) \end{cases}. \qquad (SR'_q)$$

Below are some observations on $q^*(m_i)$. First, for any $(c, \bar{q}) \in \mathbb{R}_{++} \times [0,1]$, when $\Delta_i = 1$, $q^*(m_i)$ is still well defined because $[\frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}, \frac{\Delta_i}{12}) = [\frac{1}{12}, \frac{1}{12}) = \varnothing$. Second, $\lim_{\Delta_i \downarrow 0} q^*(m_i) = 0$. Intuitively, since a message associated with some Θ_i with $\Delta_i = 0$ fully reveals the realization of the true state, upon receiving such a message, the receiver can always choose an action that exactly matches the true state regardless of her acquisition effort. Thus, she would choose to exert zero effort to maximize her payoff by minimizing her information cost.

Finally, define a function $q^0 : \mathbb{R}_{++} \times [0,1]^2 \to [0,1]$ so that for any $(c, \bar{q}, \Delta_i) \in \mathbb{R}_{++} \times [0,1]^2$,

$$q^0(c, \bar{q}, \Delta_i) := \begin{cases} q^*(m_i) & \text{if } \Delta_i \in (0,1] \\ 0 & \text{if } \Delta_i = 0 \end{cases}.$$

Accordingly, $q^0(.,.,.)$ is non-increasing in c on $\mathbb{R}_{++}$, non-decreasing in $\bar{q}$ on $[0,1]$, non-decreasing in Δ_i on $[0,1]$, and it is continuous on $(0, \frac{1}{12}) \times [0,1]^2$.

3. Stage 1 (SR_m).

In this stage, an equilibrium messaging strategy of the sender separates all types in Θ into an equilibrium partition $\{\Theta_i\}_{i \in I}$ so that it is not profitable for the sender of any type $\theta \in \Theta_j$, $j \in I$ to deviate by sending any message $m_k \in M_k$ associated with Θ_k (i.e., $M_k = \text{supp}(\theta_k)$ for all $\theta_k \in \Theta_k$) for any $k \in I \setminus \{j\}$.

First, consider the possible existence of a babbling equilibrium, where $I = \{1\}$, $\Theta_1 = \Theta = [0,1]$, and $\mu^*(\theta|m) = 1$ for all $\theta \in \Theta$ and all $m \in M$. Note that, in this case, by SR'_y, $y^*(m,s) = q^*(m)s + 0.5 \times [1 - q^*(m)]$ for all $m \in M$ and all $s \in \Theta$, while for the sender of any type $\theta \in \Theta$, his expected payoff of sending any $m \in M$ is $U(m|\theta) = -\int_0^1 [y^*(m,s) - \theta - b]^2 ds$. Accordingly, for any $\theta \in \Theta$ and any $m, m' \in M$, since $q^*(m) = q^*(m')$, $U(m|\theta) = U(m'|\theta)$, implying that SR_m is satisfied for all $\theta \in \Theta$. Hence, there always exists a babbling equilibrium.

Next, for any candidate equilibrium partition $\{\Theta_i\}_{i \in I}$ with $\#I \geq 2$, any $i \in I$, and any

$m_i \in M_i$, let $q_i := q^0(c, \bar{q}, \Delta_i)$ and

$$\begin{aligned} U_i(\theta) := U(m_i|\theta) &= -q_i[y_i(\theta) - \theta - b]^2 - (1-q_i)\int_0^1 [y_i(s) - \theta - b]^2 ds \\ &= q_i\Big\{ -\Big[\hat{\theta}_i - \theta - b - \frac{\mathbb{1}[\Delta_i > 0]\mathbb{1}[\theta \in \Theta_i] q_i(\hat{\theta}_i - \theta)}{q_i(1-\Delta_i) + \Delta_i}\Big]^2\Big\} \\ &\quad + (1-q_i)\Big\{ -(\hat{\theta}_i - \theta - b)^2 - \frac{\mathbb{1}[\Delta_i > 0] q_i^2 \Delta_i^3}{12[q_i(1-\Delta_i) + \Delta_i]^2}\Big\}. \end{aligned}$$

Then, SR_m is satisfied for all $\theta \in \Theta$ when for any $i \in I$ and any $\theta \in \Theta_i \subset \Theta$,

$$i \in \operatorname*{argmax}_{j \in I} U_j(\theta). \tag{SR'_m}$$

Proposition 1 (Sender's Finite Separation)**.** *For any equilibrium partition $\{\Theta_i\}_{i\in I}$, I is finite.*

By Proposition 1, for any equilibrium partition $\{\Theta_i\}_{i\in I}$, there is some $n \in \mathbb{N}_+$ so that the finite index set I can be written as $I = \{1, \ldots, n\}$. Next, assume without loss of generality that $\Theta_1 = [\theta_0, \theta_1]$ and $\Theta_i = (\theta_{i-1}, \theta_i]$ for all $i \in I \setminus \{1\}$ with $\theta_0 = 0$ and $\theta_n = 1$. Then, $\hat{\theta}_i = \frac{\theta_i + \theta_{i-1}}{2}$ and $\Delta_i = \theta_i - \theta_{i-1} \in (0, 1]$ for all $i \in I$. Note that, with $\#I = n \in \mathbb{N}_+$, the partition $\{\Theta_i\}_{i\in I}$ can be fully characterized by the vector of lengths of its intervals $\Delta = (\Delta_1, \ldots, \Delta_n) \in (0, 1]^n$. I will use these two representations of a partition ($\{\Theta_i\}_{i\in I}$ and Δ) interchangeably hereafter.

Proposition 2 (Sender's Sequential Rationality)**.** *For any equilibrium partition $\{\Theta_i\}_{i\in I}$ with $\#I = n \geq 2$, SR_m is satisfied for all $\theta \in \Theta$ if and only if for any $i \in I \setminus \{n\}$,*

$$D_i^+|_\Delta := \lim_{\theta \downarrow \theta_i} D_i(\theta) \geq 0 \geq \lim_{\theta \uparrow \theta_i} D_i(\theta) =: D_i^-|_\Delta,$$

where for any $\theta \in \Theta$ and any $i \in I \setminus \{n\}$, $D_i(\theta) := U_{i+1}(\theta) - U_i(\theta)$.

Proposition 2 gives necessary and sufficient conditions for SR_m to be satisfied for all types of the sender. It states that for any $i \in I$ and any $\theta \in \Theta_i$, SR_m is satisfied if and only if each cutoff type of Θ_i (including θ_{i-1} when $i \neq 1$ and θ_i when $i \neq n$) has no incentive to deviate by sending any message associated with the relevant neighboring interval (i.e., Θ_{i-1} for θ_{i-1} and Θ_{i+1} for θ_i). Its proof follows the idea of Ishida and Shimizu (2019). In their model, such no-local-deviation conditions are necessary for an equilibrium messaging strategy by definition and sufficient because, under these conditions, for any $i, j \in I$, the sender of any type $\theta \in \Theta_i$ prefers sending messages associated with Θ_j to sending those associated with Θ_{j+1} if $j \geq i$ (and those associated with Θ_{j-1} if $j \leq i$). This argument still holds in my model

though the probability q is chosen by the receiver and may differ upon receiving different messages.

Note that, when $U_i(.)$ is continuous on Θ for all $i \in I$, the conditions stated in Proposition 2 will reduce to indifference conditions for the cutoff types $\theta_1, \ldots, \theta_{n-1}$ of the partition $\Delta \in (0,1]^n$, while for any $i \in I$, by definition, $U_i(.)$ is continuous at the cutoff type(s) of Θ_i if and only if $q_i = 0$ (e.g., in the CS model) though it is continuous elsewhere for any $q_i \in [0,1]$. This is because $y_i(s)$ is discontinuous at the cutoff type(s) of Θ_i when $q_i \neq 0$, which results from the receiver's Bayesian updating in stage 3. To be more specific, when the receiver exerts positive acquisition effort and the signal s is drawn from δ_θ, the updating of the receiver's belief depends on whether type θ sender is following the equilibrium messaging strategy: if yes, the receiver will consider s to be possibly truth-revealing; otherwise, she will surely treat s as a randomly drawn noise. $y_i(s)$ is discontinuous at the cutoff type(s) of Θ_i because, in any sufficiently small neighborhood of any cutoff type θ_i^c, the types smaller than θ_i^c are following the equilibrium messaging strategy when sending any message $m_i \in M_i$ if and only if the types larger than θ_i^c are deviating from the strategy when sending any $m_i \in M_i$ (and vice versa).

SR'_y, SR'_q, and Proposition 2 fully characterize the set of equilibrium partitions with any number of intervals $n \in \mathbb{N}_+$. Define the set of n-interval equilibrium partitions under parameter values $(b, c, \bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$ as $\mathcal{E}_n(b,c,\bar{q}) \subset (0,1]^n$ for all $n \in \mathbb{N}_+$. In particular, $\mathcal{E}_1(b,c,\bar{q}) := \{(1)\}$, and for any $n \geq 2$,

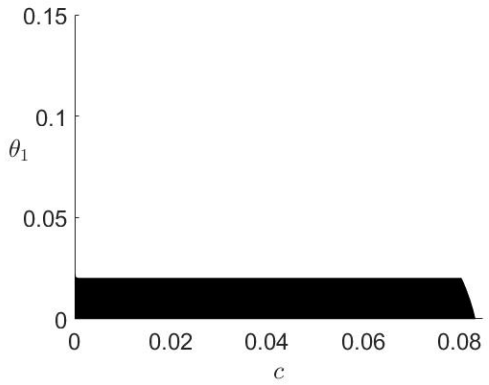
$$\mathcal{E}_n(b,c,\bar{q}) := \Big\{\Delta \in (0,1]^n : \sum_{i\in I} \Delta_i = 1, q_i = q^0(c,\bar{q},\Delta_i) \text{ for all } i \in I, \\ D_i^+|_\Delta \geq 0 \geq D_i^-|_\Delta \text{ for all } i \in I \setminus \{n\}\Big\},$$

where $I = \{1, \ldots, n\}$, and, by definition of $D_i^+|_\Delta$ and $D_i^-|_\Delta$, for any $i \in I \setminus \{n\}$,

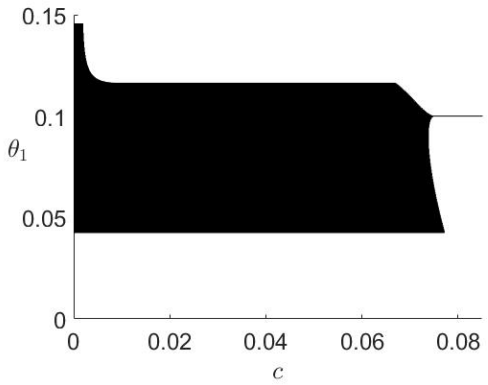
$$\begin{aligned} D_i^+|_\Delta = &-(\frac{\Delta_{i+1}}{2} - b)^2 + (\frac{\Delta_i}{2} + b)^2 + \frac{\mathbb{1}[\Delta_{i+1} > 0] q_{i+1}^2 \Delta_{i+1} (\frac{\Delta_{i+1}}{4} - b)}{q_{i+1}(1-\Delta_{i+1}) + \Delta_{i+1}} \\ &+ \frac{\mathbb{1}[\Delta_{i+1} > 0](1-q_{i+1}) q_{i+1}^2 \Delta_{i+1}^3}{6[q_{i+1}(1-\Delta_{i+1}) + \Delta_{i+1}]^2} + \frac{\mathbb{1}[\Delta_i > 0](1-q_i) q_i^2 \Delta_i^3}{12[q_i(1-\Delta_i) + \Delta_i]^2}, \\ D_i^-|_\Delta = &-(\frac{\Delta_{i+1}}{2} - b)^2 + (\frac{\Delta_i}{2} + b)^2 - \frac{\mathbb{1}[\Delta_i > 0] q_i^2 \Delta_i (\frac{\Delta_i}{4} + b)}{q_i(1-\Delta_i) + \Delta_i} \\ &- \frac{\mathbb{1}[\Delta_{i+1} > 0](1-q_{i+1}) q_{i+1}^2 \Delta_{i+1}^3}{12[q_{i+1}(1-\Delta_{i+1}) + \Delta_{i+1}]^2} - \frac{\mathbb{1}[\Delta_i > 0](1-q_i) q_i^2 \Delta_i^3}{6[q_i(1-\Delta_i) + \Delta_i]^2}. \end{aligned}$$

Note that, in general, for any $n \geq 2$, $\mathcal{E}_n(b,c,\bar{q})$ is not necessarily closed, thus not necessarily compact. This is different from the result in the CS model, where Crawford and Sobel (1982)

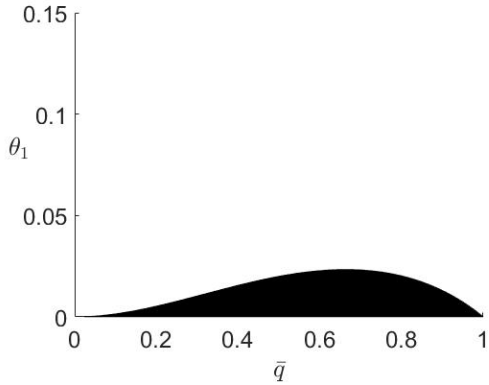
Panel A. $\mathcal{E}_2(0.25, c, 0.5)$ for all $c \in [0, 0.1]$



Panel B. $\mathcal{E}_2(0.2, c, 0.5)$ for all $c \in [0, 0.1]$



Panel C. $\mathcal{E}_2(0.25, 0.05, \bar{q})$ for all $\bar{q} \in [0, 1]$



Panel D. $\mathcal{E}_2(0.2, 0.05, \bar{q})$ for all $\bar{q} \in [0, 1]$

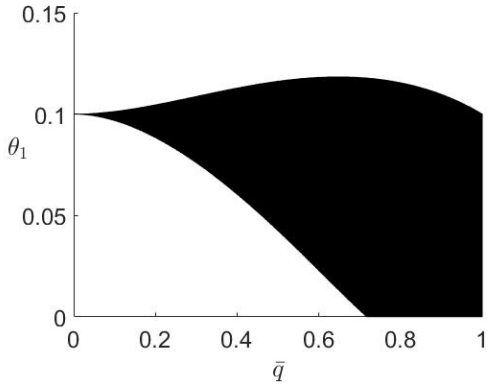


Figure 1: Cutoff Types of Two-Interval Equilibrium Partitions

have shown that the set of n-interval equilibrium partitions $\mathcal{E}_n(b, c, 0)$ is either an empty set or a singleton set for any $n \in \mathbb{N}_+$. In fact, in my model, for any $n \geq 2$, $\mathcal{E}_n(b, c, \bar{q})$ may be uncountable and contain a continuum of equilibrium partitions under a great variety of parameter values. For example, in Figure 1, each panel describes the sets of two-interval equilibrium partitions under some parameter values by plotting the value of the cutoff type (θ_1, which is also the length of the first interval Δ_1) of each partition in each set of equilibrium partitions.[2] From Figure 1, under most of the parameter values involved in the figure, there exists a continuum of equilibrium partitions.

The construction of the continuum of equilibrium partitions contained by $\mathcal{E}_n(b, c, \bar{q})$ can be based on the n-interval equilibrium partition in the CS model, which is also an equilibrium partition in my model for all $c > 0$ and all $\bar{q} \in [0, 1]$. To be more specific, for any $n \in \mathbb{N}_+$, denote an n-interval CS equilibrium partition under bias level $b > 0$ as $\Delta^{CS}(n, b)$, Crawford and Sobel (1982) showed that for any $n \in \mathbb{N}_+ \setminus \{1\}$, $\Delta^{CS}_{i+1}(n, b) = \Delta^{CS}_i(n, b) + 4b$ for all

[2]For example, panel A presents the value of the cutoff type of each equilibrium partition in $\mathcal{E}_2(0.25, c, 0.5)$ for all $c \in [0, 0.1]$.

$i \in \{1, \ldots, n-1\}$. Accordingly, for any $n \in \mathbb{N}_+ \setminus \{1\}$, $\Delta^{CS}(n,b)$ exists if and only if $b < \frac{1}{2n(n-1)}$ and is unique whenever it exists. Thus, for any $n \in \mathbb{N}_+ \setminus \{1\}$ and any $b \in (0, \frac{1}{2n(n-1)})$, $\Delta^{CS}(n,b) \in (0,1)^n \subset (0,1]^n$ and $\mathcal{E}_n(b,c,0) = \{\Delta^{CS}(n,b)\} \neq \varnothing$ for all $c > 0$.

Lemma 1 (Persistence of CS Equilibrium Partitions)**.** *For any $n \in \mathbb{N}_+ \setminus \{1\}$ and any $(b,c,\bar{q}) \in (0, \frac{1}{2n(n-1)}) \times \mathbb{R}_{++} \times [0,1]$, $\Delta^{CS}(n,b) \in \mathcal{E}_n(b,c,\bar{q})$.*

Lemma 1 formally shows that any non-babbling CS equilibrium partition is still an equilibrium partition in my model with the receiver's costly information. This is because, for any cutoff type sender, his expected payoff will be increased by the costly information acquired by the receiver if and only if he follows the messaging strategy associated with the partition. To see this, by definition of $U_i(\theta)$, given any CS equilibrium partition $\{\Theta_i\}_{i \in I}$, for any $i \in I$ and any $m_i \in M_i$, if $q_i = 0$ (e.g., in the CS model), for the sender of any type $\theta \in \Theta$, his expected payoff of sending m_i is $-(\hat{\theta}_i - \theta - b)^2$. Also, when $q_i \in (0,1]$, for the sender of any type $\theta \notin \Theta_i$ that deviates from the equilibrium messaging strategy by sending m_i, his expected payoff is $-(\hat{\theta}_i - \theta - b)^2 - \frac{(1-q_i)q_i^2\Delta_i^3}{12[q_i(1-\Delta_i)+\Delta_i]^2} \leq -(\hat{\theta}_i - \theta - b)^2$. Next, for a boundary type θ of Θ_i (i.e., some $\theta \in \{\theta_{i-1}, \theta_i\}$), when it is included in Θ_i, type θ sender's expected payoff of following the equilibrium messaging strategy by sending m_i is:

1. when $\theta = \theta_{i-1}$, $\lim\limits_{\theta \downarrow \theta_{i-1}} U_i(\theta) = \dfrac{q_i^2\Delta_i(\frac{\Delta_i}{4} - b)}{q_i(1-\Delta_i)+\Delta_i} + \dfrac{(1-q_i)q_i^2\Delta_i^3}{6[q_i(1-\Delta_i)+\Delta_i]^2} - (\dfrac{\Delta_i}{2} - b)^2$;
2. when $\theta = \theta_i$, $\lim\limits_{\theta \uparrow \theta_i} U_i(\theta) = \dfrac{q_i^2\Delta_i(\frac{\Delta_i}{4} + b)}{q_i(1-\Delta_i)+\Delta_i} + \dfrac{(1-q_i)q_i^2\Delta_i^3}{6[q_i(1-\Delta_i)+\Delta_i]^2} - (\dfrac{\Delta_i}{2} + b)^2$.

From above, for any $q_i \in (0,1]$, $\lim_{\theta \uparrow \theta_i} U_i(\theta) > -(\hat{\theta}_i - \theta_i - b)^2 = -(\frac{\Delta_i}{2} + b)^2$. Also, $\lim_{\theta \downarrow \theta_{i-1}} U_i(\theta) > -(\hat{\theta}_i - \theta_{i-1} - b)^2 = -(\frac{\Delta_i}{2} - b)^2$ for all $q_i \in (0,1]$ and all $i \in I \setminus \{1\}$ (so that θ_{i-1} is a cutoff type of the CS equilibrium partition Δ). This is because $\Delta_{i+1} = \Delta_i + 4b > 4b$ for all $i \in I \setminus \{n\}$, i.e., $\Delta_i > 4b$ for all $i \in I \setminus \{1\}$.

Corollary 1 (Uncountability of Sets of Equilibrium Partitions)**.** *For any $n \in \mathbb{N}_+ \setminus \{1\}$ and any $(b,c,\bar{q}) \in (0, \frac{1}{2n(n-1)}) \times (0, \frac{\Delta_n^{CS}(n,b)}{12}) \times (0,1]$, $\mathcal{E}_n(b,c,\bar{q})$ is uncountable.*

Corollary 1 formally states that, in my model, there exist uncountably many equilibrium partitions when the receiver exerts positive acquisition effort upon receiving some message from the sender who is following a non-babbling CS equilibrium messaging strategy. In the statement above, $b < \frac{1}{2n(n-1)}$ guarantees the existence of a CS equilibrium partition with $n \geq 2$ intervals; $\bar{q} \in (0,1]$ excludes the case with $\bar{q} = 0$ where the receiver can never exert positive effort; $c < \frac{\Delta_n^{CS}(n,b)}{12}$ is, by SR'_q, the condition for the receiver to exert positive effort upon receiving some message associated with $\Delta_n^{CS}(n,b)$, the longest interval of the CS equilibrium

partition.[3] Note that since $q^0(.,.,.)$ is non-decreasing in Δ_i on $[0,1]$, given the partition that characterizes the messaging strategy followed by the sender, the receiver (1) exerts positive (or full) effort upon receiving some message if and only if she exerts positive (or full) effort upon receiving a message associated with the longest interval of the partition and (2) exerts positive (or full) effort upon receiving every message if and only if she exerts positive (or full) effort upon receiving a message associated with the shortest interval of the partition.

4 Welfare

For any parameter values $(b,c,\bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$ and any partition Δ with index set I, let $q_i := q^0(c,\bar{q},\Delta_i)$ for all $i \in I$. Then, by definition of $U_i(\theta)$, the sender's ex-ante expected payoff (referred to as the sender's payoff hereafter) is

$$\bar{U}(b,c,\bar{q})|_\Delta := \sum_{i\in I}\int_{\theta_{i-1}}^{\theta_i} U_i(\theta) = -b^2 - \sum_{i\in I}\mathbb{1}[\Delta_i > 0]\frac{\Delta_i^3(1-q_i)(\Delta_i+q_i)}{12[q_i(1-\Delta_i)+\Delta_i]}.$$

Also, by definition of $V(q|m_i)$ (with $m_i \in M_i$), the receiver's ex-ante expected payoff (referred to as the receiver's payoff hereafter) is

$$\bar{V}(b,c,\bar{q})|_\Delta := \sum_{i\in I}\Delta_i V(q_i|m_i) = -c\sum_{i\in I}\Delta_i q_i^2 - \sum_{i\in I}\mathbb{1}[\Delta_i > 0]\frac{\Delta_i^3(1-q_i)(\Delta_i+q_i)}{12[q_i(1-\Delta_i)+\Delta_i]}.$$

Thus, define the ex-ante social welfare (referred to as welfare hereafter) as

$$\begin{aligned} W^\alpha(b,c,\bar{q})|_\Delta :=& \alpha\bar{U}(b,c,\bar{q})|_\Delta + (1-\alpha)\bar{V}(b,c,\bar{q})|_\Delta \\ =& -\alpha b^2 - (1-\alpha)c\sum_{i\in I}\Delta_i q_i^2 - \sum_{i\in I}\mathbb{1}[\Delta_i > 0]\frac{\Delta_i^3(1-q_i)(\Delta_i+q_i)}{12[q_i(1-\Delta_i)+\Delta_i]}, \end{aligned}$$

where $\alpha \in [0,1]$ is the social welfare weight on the sender.

Recall from the previous section that $\mathcal{E}_n(b,c,\bar{q}) \subset (0,1]^n$ is not necessarily compact for any $n \geq 2$. Denote the closure of $\mathcal{E}_n(b,c,\bar{q})$ as $\bar{\mathcal{E}}_n(b,c,\bar{q}) \subset [0,1]^n$. Note that $\mathcal{E}_1(b,c,\bar{q}) = \bar{\mathcal{E}}_1(b,c,\bar{q}) = \{(1)\}$ for all $(b,c,\bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$.

Also, for any $(b,c,\bar{q}) \in \mathbb{R}_{++} \times [\frac{1}{12},\infty) \times [0,1]$, by SR'_q, $q^0(c,\bar{q},\Delta_i) = 0$ for all $\Delta_i \in [0,1]$. Thus, for any $n \geq 2$, by definition of $\mathcal{E}_n(b,c,\bar{q})$, $\Delta \in \mathcal{E}_n(b,c,\bar{q})$ if and only if $\Delta_{i+1} = \Delta_i + 4b$ for all $i \in \{1,\ldots,n-1\}$. Accordingly, for any $n \in \mathbb{N}_+$, $\bar{\mathcal{E}}_n(b,c,\bar{q}) = \mathcal{E}_n(b,c,\bar{q}) = \mathcal{E}_n(b,c,0)$, which is either a singleton set or empty. Thus, in this case, $\max_{\Delta\in\bar{\mathcal{E}}_n(b,c,\bar{q})} W^\alpha(b,c,\bar{q})|_\Delta$ is well defined whenever $\bar{\mathcal{E}}_n(b,c,\bar{q}) \neq \varnothing$.

[3] $\Delta_n^{CS}(n,b)$ is the longest interval of $\Delta^{CS}(n,b)$ because $\Delta_{i+1}^{CS}(n,b) = \Delta_i^{CS}(n,b) + 4b > \Delta_i^{CS}(n,b)$ for any $i \in \{1,\ldots,n-1\}$.

Next, for any $(b,c,\bar{q}) \in \mathbb{R}_{++} \times (0, \frac{1}{12}) \times [0,1]$ and any $n \in \mathbb{N}_+$, by definition of $W^\alpha(b,c,\bar{q})|_\Delta$, since $q^0(.,.,.)$ is continuous on $(0, \frac{1}{12}) \times [0,1]^2$, $W^\alpha(b,c,\bar{q})|_\Delta$ is continuous in Δ on $[0,1]^n$. Thus, for any $(b,c,\bar{q}) \in \mathbb{R}_{++} \times (0, \frac{1}{12}) \times [0,1]$ and any $n \in \mathbb{N}_+$ with $\bar{\mathcal{E}}_n(b,c,\bar{q}) \neq \varnothing$, $\max_{\Delta \in \bar{\mathcal{E}}_n(b,c,\bar{q})} W^\alpha|_\Delta$ is well defined because the continuous function $W^\alpha|_\Delta$ attains its maximum over the compact set $\bar{\mathcal{E}}_n(b,c,\bar{q})$.

Then, for any $(b,c,\bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$, let $\mathcal{N}(b,c,\bar{q}) := \{n \in \mathbb{N}_+ : \mathcal{E}_n(b,c,\bar{q}) \neq \varnothing\}$. Note that since $\mathcal{E}_1(b,c,\bar{q}) = \{(1)\} \neq \varnothing$, $1 \in \mathcal{N}(b,c,\bar{q})$, implying $\mathcal{N}(b,c,\bar{q}) \neq \varnothing$. Also, by Proposition 1, $\mathcal{N}(b,c,\bar{q})$ is a finite set. Accordingly, the supremum of the equilibrium welfare under parameter values $(b,c,\bar{q})$ and welfare weight α is

$$\begin{aligned}\bar{W}^\alpha(b,c,\bar{q}) &:= \max_{n \in \mathcal{N}(b,c,\bar{q})} \sup_{\Delta \in \mathcal{E}_n(b,c,\bar{q})} W^\alpha(b,c,\bar{q})|_\Delta \\ &= \max_{n \in \mathcal{N}(b,c,\bar{q})} \max_{\Delta \in \bar{\mathcal{E}}_n(b,c,\bar{q})} W^\alpha(b,c,\bar{q})|_\Delta,\end{aligned}$$

which is well defined for all $(b,c,\bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$ and all $\alpha \in [0,1]$.

Finally, since the comparative static analysis in the following sections concentrates on the maximum (rather than supremum) value of the equilibrium welfare under different parameter values, it is necessary to show that the maximum equilibrium welfare is always well defined to justify the analysis.

Lemma 2 (Existence of Maximum Equilibrium Welfare)**.** *For any $(b,c,\bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$ and any $\alpha \in [0,1]$, there exists some $n \in \mathcal{N}(b,c,\bar{q})$ and some $\Delta \in \mathcal{E}_n(b,c,\bar{q})$ so that $W^\alpha(b,c,\bar{q})|_\Delta = \bar{W}^\alpha(b,c,\bar{q})$.*

By Lemma 2, for any parameter values $(b,c,\bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$ and any welfare weight $\alpha \in [0,1]$, there exists some equilibrium partition whose associated welfare level attains $\bar{W}^\alpha(b,c,\bar{q})$, the supremum of the welfare function over the set of equilibrium partitions, which is therefore the maximum equilibrium welfare. The intuition is that, for any "partition" that is in the closure of the set of equilibrium partitions but is not an equilibrium partition (where the only possible case is that it contains some zero-measure "intervals", i.e., discrete points), there always exists an equilibrium partition that leads to the same welfare level as it does, and the equilibrium partition can be constructed simply by letting the types represented by the discrete points pool with the types in their respective neighboring intervals.

4.1 Direct Effect

In general, it is technically difficult to explicitly characterize an equilibrium partition that maximizes welfare due to the mathematical complexity associated with the set of equilibrium

partitions and the social welfare function. Instead, I analyze the total effect of a change in the costly information source on the maximum equilibrium welfare by decomposing it into two parts: direct effect and indirect effect. The direct effect holds the sender's messaging strategy (i.e., the partition) fixed and explains how the consequent changes in the receiver's effort level and action directly affect the maximum equilibrium welfare. With the partition held fixed, the problem reduces to a decision problem faced by the receiver, who chooses her effort level and action sequentially and affects the welfare. The indirect effect allows for reoptimization over the new set of equilibrium partitions (under the new parameter values after the change) and involves the complexity associated with the strategic interactions between the two parties. This section focuses on the direct effect, and the next section will concentrate on the total effect, the analysis of which will be on the basis of the results of the direct effect and circumvent some technical difficulties associated with the indirect effect.

Note that, by definition of social welfare, given the bias level $b > 0$ and the welfare weight $\alpha \in [0,1]$, for a change in the information source with parameter values changed from some $(c, \bar{q}) \in \mathbb{R}_{++} \times [0,1]$ to some $(c', \bar{q}') \in \mathbb{R}_{++} \times [0,1]$, the direct effect of the change conditional on some partition Δ is $W^\alpha(b, c', \bar{q}')|_\Delta - W^\alpha(b, c, \bar{q})|_\Delta$, and the total effect of the change is $\bar{W}^\alpha(b, c', \bar{q}') - \bar{W}^\alpha(b, c, \bar{q})$.

Proposition 3 (Direct Effects of Changes in the Information Source)**.** *For any $b, c, c' \in \mathbb{R}_{++}$ so that $c > c'$, any $\bar{q}, \bar{q}' \in [0,1]$ so that $\bar{q} < \bar{q}'$, any $\alpha \in [0,1]$, any $n \in \mathbb{N}_+$, and any partition $\Delta \in (0,1]^n$,*

1. *$W^\alpha(b, c', \bar{q})|_\Delta - W^\alpha(b, c, \bar{q})|_\Delta \geq 0$, and the inequality is strict if and only if $\bar{q} > 0$, $c' < \frac{\Delta_M}{12}$, and, when $c \leq \frac{\Delta_m^2[\bar{q}(1-\Delta_m)+2\Delta_m]}{24[\bar{q}(1-\Delta_m)+\Delta_m]^2}$, $\alpha < 1$;*
2. *$W^\alpha(b, c, \bar{q}')|_\Delta - W^\alpha(b, c, \bar{q})|_\Delta \geq 0$, and the inequality is strict if and only if $c < \frac{\Delta_M^2[\bar{q}(1-\Delta_M)+2\Delta_M]}{24[\bar{q}(1-\Delta_M)+\Delta_M]^2}$;*

where $\Delta_M = \max\{\Delta_i\}_{i \in I}$, $\Delta_m = \min\{\Delta_i\}_{i \in I}$, and $I = \{1, \ldots, n\}$ is the index set of Δ.

Proposition 3 is about the direct effect of a reduction in the cost of information acquisition and that of an increase in the informativeness of the costly information source. These two types of changes are respectively characterized by a decrease in the cost parameter c and an increase in the informativeness parameter $\bar{q}$. Part 1 is about the former, and part 2 is about the latter. Proposition 3 shows that the direct effect of each type of change is always non-negative and specifies the necessary and sufficient conditions for a positive direct effect in each case. Both parts of it indicate that the direct effect is positive if and only if the receiver is induced to exert greater acquisition effort upon receiving some message (or, for a cost reduction, if and only if the receiver's information cost is lower when the reduction does not affect the effort level). An increase in informativeness induces the receiver to exert

greater effort if and only if, before the change, the receiver would exert greater effort upon receiving some message if she were not constrained by the less informative information source. Otherwise, the increase in informativeness will not affect the receiver's effort level. A cost reduction increases the receiver's effort level if and only if the receiver exerts positive effort upon receiving some message after the reduction (which requires the information source to be informative, i.e., $\bar{q} > 0$). Also, if the cost reduction does not affect the receiver's effort level, it still directly increases welfare when the receiver exerts full effort (i.e., $q = \bar{q}$) upon receiving every message even before the reduction. In this case, the cost reduction only benefits by reducing the receiver's information cost.

Note that the direct effect is conditional on an arbitrarily fixed partition in Proposition 3 though the comparative static analysis in this paper is about the maximum equilibrium welfare. This is because there may not be a unique equilibrium maximizer and, in the analysis, when the set of equilibrium welfare maximizers is not a singleton set, there is no perfect criterion for choosing an equilibrium welfare maximizer which the direct effect will be conditional on (especially given that the set of equilibrium welfare maximizers is not explicitly characterized). Also, as a result, Proposition 3 shows that the direct effect conditional on any partition is non-negative (and it is positive under some conditions related to the partition), which implies that the direct effect conditional on any equilibrium welfare maximizer is non-negative (and it is positive under the corresponding conditions related to the maximizer).

4.2 Total Effect

By definition, the total effect of a change in the costly information source is the sum of the direct effect and the indirect effect. In general, the sign of the indirect effect is unclear when the equilibrium welfare maximizers under different parameter values are not explicitly characterized. Since the direct effect is non-negative, a non-negative indirect effect implies a non-negative total effect, while the sign of the total effect is unclear when the indirect effect is negative.

There exist cases where the indirect effect of a change is always non-negative. In particular, one condition for the indirect effect to be always non-negative is that an equilibrium partition that maximizes welfare before the change is still an equilibrium partition after the change. In this case, the reoptimization over the new set of equilibrium partitions never decreases welfare because it can stick to the equilibrium welfare maximizer before the change to keep the welfare unchanged.

Corollary 2 (Implications of Non-negative Indirect Effects for Total Effects)**.** *For any*

$b, c, c' \in \mathbb{R}_{++}$, so that $c > c'$, any $\bar{q}, \bar{q}' \in [0,1]$ so that $\bar{q} < \bar{q}'$, any $\alpha \in [0,1]$, any $n \in \mathcal{N}(b,c,\bar{q})$, and any partition $\Delta \in \mathcal{E}_n(b,c,\bar{q})$ so that $W^\alpha(b,c,\bar{q})|_\Delta = \bar{W}^\alpha(b,c,\bar{q})$:

1. *if $\Delta \in \mathcal{E}_n(b,c',\bar{q})$, then $\bar{W}^\alpha(b,c',\bar{q}) - \bar{W}^\alpha(b,c,\bar{q}) \geq 0$, and the inequality is strict if $\bar{q} > 0$, $c' < \frac{\Delta_M}{12}$, and $\alpha < 1$;*

2. *if $\Delta \in \mathcal{E}_n(b,c,\bar{q}')$, then $\bar{W}^\alpha(b,c,\bar{q}') - \bar{W}^\alpha(b,c,\bar{q}) \geq 0$, and the inequality is strict if $c < \frac{\Delta_M^2[\bar{q}(1-\Delta_M)+2\Delta_M]}{24[\bar{q}(1-\Delta_M)+\Delta_M]^2}$.*

Corollary 2 immediately follows from Proposition 3 and reflects the fact that when the indirect effect of a change is non-negative, the total effect of the change is non-negative (or positive) if the direct effect of it is non-negative (or positive). Note that, when the corresponding indirect effect is non-negative, a positive direct effect is sufficient but not necessary for a positive total effect because there may exist cases where the direct effect is zero while the indirect effect is positive.

4.2.1 Effect of Access to an Informative Information Source

The case where the receiver in the CS model obtains access to an informative costly information source is characterized by an increase in the informativeness parameter $\bar{q}$ from 0 to some positive value $\bar{q}' \in (0,1]$. Note that, by Lemma 1, each CS equilibrium partition is still an equilibrium partition in my model under the same bias level b regardless of the values of the other parameters (c and $\bar{q}$). Therefore, in this case, the indirect effect is always non-negative.

Proposition 4 (Total Effect of Access)**.** *For any $(b,c,\bar{q}') \in \mathbb{R}^2_{++} \times (0,1]$ and any $\alpha \in [0,1]$,*

$$\bar{W}^\alpha(b,c,\bar{q}') - \bar{W}^\alpha(b,c,0) \geq 0,$$

and the inequality is strict if $c < \frac{\Delta_M^{CS}(b)}{12}$, where $\Delta_M^{CS}(b) := \frac{1}{N^{CS}(b)} + 2[N^{CS}(b) - 1]b$ and $N^{CS}(b) := \begin{cases} \max\{n \in \mathbb{N}_+ : 2n(n-1)b < 1\} & \text{if } b < \frac{1}{4} \\ 1 & \text{else} \end{cases}$.

Proposition 4 indicates that the total effect of the receiver's access to an informative costly information source is always non-negative. This is a direct implication of a non-negative direct effect (by Proposition 3) and a non-negative indirect effect (by Lemma 1). Also, the total effect is positive if the cost of information acquisition is sufficiently low so that, after the receiver obtains access to the information source, she would exert positive effort upon receiving some message under the messaging strategy associated with $\Delta^{CS}[N^{CS}(b), b]$, the

CS equilibrium partition with the most intervals under bias level b. $N^{CS}(b)$ is derived from the equilibrium condition of the CS model. The sufficient condition for a positive total effect involves $\Delta^{CS}[N^{CS}(b), b]$ because this partition is the equilibrium welfare maximizer in the CS model under bias level b. To be more specific, Crawford and Sobel (1982) showed that, in the CS model, both the sender and the receiver strictly prefer $\Delta^{CS}[N^{CS}(b), b]$ to any other equilibrium partition with fewer intervals. Thus, in the CS model, the associated welfare level of $\Delta^{CS}[N^{CS}(b), b]$ is higher than that of any other equilibrium partition regardless of the welfare weights.

4.2.2 Effects of Cost Reduction and Increase in Informativeness

This section focuses on the total effect of a reduction in the cost of information acquisition and that of an increase in the informativeness of the costly information source. The previous case about access is a special case of an increase in informativeness. In general, an equilibrium welfare maximizer before a change in the information source need not be an equilibrium partition after the change. By Proposition 2, a partition is supported by an equilibrium if each of its cutoff types has no incentive to locally deviate by pooling with the types in the corresponding neighboring interval. Thus, an equilibrium partition before a change remains to be an equilibrium partition after the change if the change never strengthens any cutoff type sender's incentive to locally deviate. A sufficient condition for this is that the change weakly increases all cutoff type senders' expected payoffs of following the messaging strategy associated with the partition and, on the contrary, weakly decreases those of locally deviating from that messaging strategy.

Lemma 3 (Sufficient Conditions for Non-negative Indirect Effects)**.** *For any $b \in \mathbb{R}_{++}$, any $c, c' \in [\frac{b}{3}, +\infty)$ so that $c > c'$, any $\bar{q}, \bar{q}' \in [0, \bar{\bar{q}}]$ so that $\bar{q} < \bar{q}'$, any $n \in \mathcal{N}(b, c, \bar{q})$, and any partition $\Delta \in \mathcal{E}_n(b, c, \bar{q})$,*

$$\Delta \in \mathcal{E}_n(b, c', \bar{q}) \bigcap \mathcal{E}_n(b, c, \bar{q}'),$$

where $\bar{\bar{q}} \in [0, 1]$ is defined by $\bar{\bar{q}}^2(1 - 4b) + 12b\bar{\bar{q}} = 8b$.

Lemma 3 formally states a set of sufficient conditions for a change in the costly information source to always incentivize the obedience of and discourage the deviation of every cutoff type sender in any equilibrium partition before the change. In particular, the conditions are that the cost of information acquisition is not too low and the information source is not too informative. The lower bound of c ($\frac{b}{3}$) guarantees that the receiver exerts zero effort upon receiving any message associated with an interval no longer than $4b$. As a result, any cutoff type sender's payoff of sending any message associated with such a short interval is invariant

to any change in the information source. The upper bound of $\bar{q}$ ($\bar{\bar{q}}$, which has a unique value on $[0,1]$ for each $b \in \mathbb{R}_{++}$) is related to intervals longer than $4b$. Both types of change in the information source weakly increase the receiver's effort level upon receiving every message, and the upper bound $\bar{\bar{q}}$ guarantees that, conditional on any equilibrium partition before a change, for the sender of any cutoff type of the partition who sends a message associated with an interval longer than $4b$, any such increase in the receiver's effort level would (1) weakly increase his payoff if he is following the messaging strategy associated with the partition and (2) weakly decrease his payoff if he is deviating from that messaging strategy.

Proposition 5 (Total Effect of Cost Reduction)**.** *For any $b \in \mathbb{R}_{++}$, any $c, c' \in [\frac{b}{3}, +\infty)$ so that $c > c'$, any $\bar{q} \in (0, \bar{\bar{q}}]$, and any $\alpha \in [0,1]$,*

$$\bar{W}^{\alpha}(b, c', \bar{q}) - \bar{W}^{\alpha}(b, c, \bar{q}) \geq 0,$$

and the inequality is strict if $c' < \frac{\Delta_M^{CS}(b)}{12}$ and $\alpha < 1$.

Proposition 6 (Total Effect of Increase in Informativeness)**.** *For any $b \in \mathbb{R}_{++}$, any $c \geq \frac{b}{3}$, any $\bar{q}, \bar{q}' \in (0, \bar{\bar{q}}]$ so that $\bar{q} < \bar{q}'$, and any $\alpha \in [0,1]$,*

$$\bar{W}^{\alpha}(b, c, \bar{q}') - \bar{W}^{\alpha}(b, c, \bar{q}) \geq 0.$$

Proposition 5 describes sufficient conditions for the total welfare effect of a cost reduction to be non-negative and positive, respectively. In particular, conditional on a non-negative indirect effect, a cost reduction increases welfare if it either induces the receiver to exert greater effort upon receiving some message after the reduction or reduces her information cost when it does not affect her effort level. Both of the two cases need the reduced cost of information acquisition to be sufficiently low (a sufficient condition of which is $c' < \frac{\Delta_M^{CS}(b)}{12}$)[4] so that the receiver would exert positive effort. The latter case also needs the welfare weight on the receiver to be positive so that any reduction in the information cost would result in a welfare improvement. Proposition 6 states sufficient conditions for the total welfare effect of an increase in informativeness to be non-negative.

Note that Proposition 6 does not give any sufficient condition for an increase in the informativeness of the information source to induce a welfare improvement. In fact, a cost reduction generates welfare gains under broader conditions than an increase in informativeness does as long as the welfare weight on the receiver is positive. This is because, by Proposition 3, a positive direct effect of an increase in informativeness requires a lower cost of information acquisition ($c < \frac{\Delta_M^2[\bar{q}(1-\Delta_M)+2\Delta_M]}{24[\bar{q}(1-\Delta_M)+\Delta_M]^2}$) than that of a cost reduction does ($c' < \frac{\Delta_M}{12}$),

[4]See in detail in Lemma 4 (in the proof of Proposition 5) in Appendix A.

while, for the two types of changes, the remaining condition for a positive direct effect and the conditions for a non-negative indirect effect are respectively identical. In other words, a cost reduction induces a welfare improvement whenever an increase in informativeness does whereas, in general, there exist cases where a cost reduction induces a welfare improvement while an increase in informativeness does not.

5 Conclusion

This paper analyzes a cheap-talk model with a receiver who can acquire information from a costly information source. The analysis focuses on the welfare effects of two types of changes in the information source: a reduction in the cost of information acquisition and an increase in the informativeness of the information source. In general, regardless of the welfare weights on the two parties, neither type of change decreases welfare when the cost of information acquisition is not too low and the information source is not too informative. Especially, the welfare is never decreased when the informativeness is increased from zero, i.e., when the receiver obtains access to an informative information source. Also, a change increases welfare if it directly induces the receiver to exert greater effort upon receiving some message from the sender, and, in general, a cost reduction induces a welfare improvement in more cases than an increase in informativeness does.

The theoretical findings of this paper can be applied in certain situations, where reducing receivers' costs of information acquisition may be considered as a better way of inducing a welfare improvement than increasing the informativeness of informative information sources. For example, in China, a local government often hopes to disclose information about its financial status (e.g., historical data stored on official websites) so as to incentivize investors to purchase more local treasury bonds. It may find that reducing investors' searching costs (e.g., improving the user interfaces of the websites) is more effective than providing more and better information (e.g., keeping the historical data undeleted for a longer period of time) in both increasing the sales of the bonds and informing investors about the bonds. This is because the former can benefit by reducing the cost of or increasing the amount of information acquired by every investor who would search for relevant information, while the latter can benefit only by providing more information to investors, many of whom are indifferent between a richer set of information and a smaller one because they do not frequently and intensively refer to and analyze the data even before a greater amount of data is provided.

However, the costly information source available to the receiver need not be so simple (as it is in my model) that the precision of the acquired costly information only depends on the receiver's acquisition effort. For instance, the sender may also be able to affect the costly

information source (e.g., in the case of customer reviews, a retailer may buy fake reviews to deceive customers). This is an interesting direction in which cheap-talk models could be further extended to study the strategic interactions between the two parties in two aspects: that between information transmission by the sender and costly information acquisition by the receiver, and that between the sender's effects on the costly information source and the receiver's interpretation of her acquired costly information.

A Omitted Proofs

A.1 Proof of Proposition 1

Claim 1. *For any equilibrium partition* $\{\Theta_i\}_{i\in I}$ *and any* $i \in I$ *so that* $0 < \underline{\theta}_i < \bar{\theta}_i$, $\bar{\theta}_i - \underline{\theta}_i \geq 2b$.

Proof of Claim 1 (Following De Barreda (2010))

Suppose not, i.e., there exists $i \in I$ so that $0 < \underline{\theta}_i < \bar{\theta}_i$ and $\bar{\theta}_i - \underline{\theta}_i < 2b$.

Thus, $|\bar{\theta}_i - \underline{\theta}_i - b| < b = |\underline{\theta}_i - \underline{\theta}_i - b|$, implying $u(\bar{\theta}_i, \underline{\theta}_i, b) > u(\underline{\theta}_i, \underline{\theta}_i, b)$.

Then, by continuity of $u(.,.,.)$, there exists some small $\epsilon \in (0, \min\{b, \underline{\theta}_i\})$ so that for $\theta' := \underline{\theta}_i - \epsilon \in (0, \underline{\theta}_i)$, $u(\bar{\theta}_i, \theta', b) > u(\underline{\theta}_i, \theta', b)$.

Next, since $u(y, \theta, b)$ is concave in y for all $(\theta, b) \in \mathbb{R}^2_{++}$, for any $\hat{\theta} = t\underline{\theta}_i + (1-t)\bar{\theta}_i$ and any $t \in (0,1)$,

$$\begin{aligned} u(\hat{\theta}, \theta', b) &\geq tu(\underline{\theta}_i, \theta', b) + (1-t)u(\bar{\theta}_i, \theta', b) \\ &> tu(\underline{\theta}_i, \theta', b) + (1-t)u(\underline{\theta}_i, \theta', b) = u(\underline{\theta}_i, \theta', b). \end{aligned}$$

Also, for any $\tilde{\theta} < \underline{\theta}_i$, $|\tilde{\theta} - \theta' - b| = b + \theta' - \tilde{\theta} = \underline{\theta}_i - \tilde{\theta} + b - \epsilon > b - \epsilon = |\underline{\theta}_i - \theta' - b|$, implying $u(\underline{\theta}_i, \theta', b) > u(\tilde{\theta}, \theta', b)$.

Finally, since $\theta' = \underline{\theta}_i - \epsilon \in (0, \underline{\theta}_i)$, there exists $j \in I$ so that $\theta' \in \Theta_j$ and $\bar{\theta}_j \leq \underline{\theta}_i$.

Thus, for any $\theta'' \in \Theta_i$ and any $\theta''' \in \Theta_j$, since $\theta'' \in [\underline{\theta}_i, \bar{\theta}_i]$ and $\theta''' \leq \bar{\theta}_j$, $u(\theta''', \theta', b) \leq u(\underline{\theta}_i, \theta', b) \leq u(\theta'', \theta', b)$, and $u(\theta''', \theta', b) = u(\theta'', \theta', b)$ only if $\theta''' = \underline{\theta}_i = \theta''$. Note that, by definition of partition, $\Theta_i \cap \Theta_j = \varnothing$, implying $\theta''' \neq \theta''$.

Therefore, for any $s, s' \in \Theta$, since SR'_y indicates that $y_i(s) \in \Theta_i$ and $y_j(s') \in \Theta_j$, $u[y_j(s'), \theta', b] < u[y_i(s), \theta', b]$. Hence, when $q_i \leq q_j$,

$$\begin{aligned} U_j(\theta') &= [q_i + (q_j - q_i)]u[y_j(\theta'), \theta', b] + (1 - q_j)\int_0^1 u[y_j(s), \theta', b]ds \\ &< q_i u[y_i(\theta'), \theta', b] + [(q_j - q_i) + (1 - q_j)]\int_0^1 u[y_i(s), \theta', b]ds = U_i(\theta'), \end{aligned}$$

and when $q_i > q_j$,

$$U_j(\theta') = q_j u[y_j(\theta'), \theta', b] + [(1-q_i) + (q_i - q_j)] \int_0^1 u[y_j(s), \theta', b] ds$$
$$< [q_j + (q_i - q_j)] u[y_i(\theta'), \theta', b] + (1-q_i) \int_0^1 u[y_i(s), \theta', b] ds = U_i(\theta').$$

Each of the above cases yields a contradiction against SR_m for $\theta = \theta' \in \Theta_j$. □

Thus, by Claim 1, for any $i \in I \setminus \{1\}$, since $\bar{\theta}_i > \underline{\theta}_i > 0$, $\bar{\theta}_i - \underline{\theta}_i \geq 2b$.
Hence, $\#I < \frac{1}{2b} + 1$, implying that I is a finite set. □

A.2 Proof of Proposition 2 (Following Ishida and Shimizu (2019))

Only if.

First, note that $n \geq 2$, $\Theta_{i+1} = (\theta_i, \theta_{i+1}]$ for all $i \in I \setminus \{n\}$, and $\Theta_1 = [\theta_0, \theta_1]$.

Also, for any $\theta \in \Theta$ and any $i \in I \setminus \{n\}$, by definition of $D_i(\theta)$, $D_i(\theta) \geq 0$ if and only if type θ sender has no incentive to deviate from sending any $m_{i+1} \in M_{i+1}$ to sending any $m_i \in M_i$, and $D_i(\theta) \leq 0$ if and only if type θ sender has no incentive to deviate from sending $m_i \in M_i$ to sending $m_{i+1} \in M_{i+1}$. Thus, SR_m requires that $D_i(\theta) \geq 0$ for all $\theta \in \Theta_{i+1}$ and $D_i(\theta) \leq 0$ for all $\theta \in \Theta_i$.

Then, for any $i \in I \setminus \{n\}$ and any $\theta \in (\theta_i, \theta_{i+1})$, by definition of $D_i(\theta)$, $D_i'(\theta) = \frac{2q_{i+1}^2\{[q_{i+1}(1-2\Delta_{i+1})+2\Delta_{i+1}](\theta-\hat{\theta}_{i+1})+b[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]\}}{[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]^2} + 2(\hat{\theta}_{i+1} - \hat{\theta}_i)$ and $D_i''(\theta) = \frac{2q_{i+1}^2[q_{i+1}(1-2\Delta_{i+1})+2\Delta_{i+1}]}{[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]^2} \geq 0$ (because $q_{i+1} \in [0,1]$ and $\Delta_{i+1} \in (0,1)$).

Thus, by continuity of $D_i'(\theta)$ on (θ_i, θ_{i+1}), the infimum of $D_i'(\theta)$ on (θ_i, θ_{i+1}) is $\lim_{\theta \downarrow \theta_i} D_i'(\theta) = -\frac{2q_{i+1}^2\{[q_{i+1}(1-2\Delta_{i+1})+2\Delta_{i+1}]\frac{\Delta_{i+1}}{2} - b[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]\}}{[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]^2} + 2(\hat{\theta}_{i+1} - \hat{\theta}_i)$.

For now, denote q_{i+1} as q and Δ_{i+1} as Δ'. Then, denote $\lim_{\theta \downarrow \theta_i} D_i'(\theta) = 2G(q)$, where $G(q) := \hat{\theta}_{i+1} - \hat{\theta}_i - \frac{q^2\{[q[(\Delta'-b)(1-\Delta')-\frac{\Delta'}{2}]+\Delta'(\Delta'-b)\}}{[q(1-\Delta')+\Delta']^2}$ for all $q \in [0,1]$. Note that $G(0) = \hat{\theta}_{i+1} - \hat{\theta}_i > 0$, $G(1) = b + \frac{\Delta_i}{2} > 0$.

Thus, $G'(q) = -\frac{q\{q[q(1-\Delta')+3\Delta'][(\Delta'-b)(1-\Delta')-\frac{\Delta'}{2}]+2\Delta'^2(\Delta'-b)\}}{[q(1-\Delta')+\Delta']^3}$, where $G'(0) = 0$ and $G'(1) = b(1+\Delta') - \frac{\Delta'}{2}$.

Then, let $H(q) := -q[q(1-\Delta')+3\Delta'][(\Delta'-b)(1-\Delta')-\frac{\Delta'}{2}] - 2\Delta'^2(\Delta'-b)$ for all $q \in [0,1]$. Note that $H(1) = G'(1) = b(1+\Delta') - \frac{\Delta'}{2}$, $H(0) = -2\Delta'^2(\Delta'-b) < 0$ (because, by Claim 1 in the proof of Proposition 1, $\Delta' \geq 2b > 0$), and for any $q \in [0,1]$, $G'(q) = \frac{qH(q)}{[q(1-\Delta')+\Delta']^3}$ and $H'(q) = -[2q(1-\Delta')+3\Delta'][(\Delta'-b)(1-\Delta')-\frac{\Delta'}{2}]$.

Case 1: $(\Delta'-b)(1-\Delta') - \frac{\Delta'}{2} \geq 0$, implying $H'(q) \leq 0$ on $[0,1]$.

Thus, $H(q) \leq H(0) < 0$ on $[0,1]$, indicating $G'(q) \leq 0$ on $[0,1]$. Therefore, for any $q \in [0,1]$, $G(q) \geq G(1) > 0$.

Case 2: $(\Delta' - b)(1-\Delta') - \frac{\Delta'}{2} < 0$, yielding $H'(q) > 0$ on $[0,1]$.

Then, if $H(1) = b(1+\Delta') - \frac{\Delta'}{2} \le 0$, then $H(q) \le H(1) \le 0$ on $[0,1]$, implying $G'(q) \le 0$ on $[0,1]$. Therefore, for any $q \in [0,1]$, $G(q) \ge G(1) > 0$.

Finally, if $H(1) = b(1+\Delta') - \frac{\Delta'}{2} > 0$, there exists $\hat{q} \in (0,1)$ so that $H(\hat{q}) = 0$, $H(q) > 0$ on $(\hat{q}, 1]$, and $H(q) < 0$ on $[0, \hat{q})$, implying that $G'(\hat{q}) = G'(0) = 0$, $G'(q) > 0$ on $(\hat{q}, 1]$, and $G'(q) < 0$ on $(0, \hat{q})$. As a result, $\min_{q\in[0,1]} G(q) = G(\hat{q})$, and $\hat{q}[(\Delta'-b)(1-\Delta') - \frac{\Delta'}{2}] + \Delta'(\Delta'-b) = \frac{\hat{q}}{2\Delta'}[\hat{q}(1-\Delta')+\Delta'][\frac{\Delta'}{2} - (\Delta'-b)(1-\Delta')]$ because $H(\hat{q}) = 0$. Therefore, with $\Delta' > 0$ and $\hat{q} \in [0,1]$,

$$
\begin{aligned}
G(\hat{q}) &= \hat{\theta}_{i+1} - \hat{\theta}_i - \hat{q}^2\{\hat{q}[(\Delta'-b)(1-\Delta') - 0.5\Delta'] + \Delta'(\Delta'-b)\}[\hat{q}(1-\Delta')+\Delta']^{-2} \\
&= \hat{\theta}_{i+1} - \hat{\theta}_i - \frac{\hat{q}^3[0.5\Delta' - (\Delta'-b)(1-\Delta')]}{2\Delta'[\hat{q}(1-\Delta')+\Delta']} \text{ (decreasing in } \hat{q} \text{ on } [0,1]) \\
&\ge \hat{\theta}_{i+1} - \hat{\theta}_i - [0.5\Delta' - (\Delta'-b)(1-\Delta')](2\Delta')^{-1} \text{ (decreasing in } b) \\
&\ge \hat{\theta}_{i+1} - \hat{\theta}_i - [0.5\Delta' - (\Delta'-0.5\Delta')(1-\Delta')](2\Delta')^{-1} \\
&\quad \text{(because, by Claim 1 in the proof of Proposition 1, } b \le 0.5\Delta') \\
&= \hat{\theta}_{i+1} - \hat{\theta}_i - 0.25\Delta' = 0.25\Delta_{i+1} + 0.5\Delta_i > 0, \text{ implying } G(q) \ge G(\hat{q}) > 0 \text{ on } [0,1].
\end{aligned}
$$

Thus, in any case, for any $\theta \in (\theta_i, \theta_{i+1})$, $D_i'(\theta) \ge \lim_{\theta\downarrow\theta_i} D_i'(\theta) = 2G(q) \ge 0$.

Accordingly, since $D_i(\theta)$ is continuous on (θ_i, θ_{i+1}) and $D_i(\theta_{i+1}) = \lim_{\theta\uparrow\theta_{i+1}} D_i(\theta)$, $\lim_{\theta\downarrow\theta_i} D_i(\theta) = \inf_{\theta\in\Theta_{i+1}} D_i(\theta)$.

Hence, $D_i(\theta) \ge 0$ for all $\theta \in \Theta_{i+1}$ if and only if $D_i^+|_\Delta = \lim_{\theta\downarrow\theta_i} D_i(\theta) \ge 0$.

Similarly, for any $i \in I \setminus \{n\}$ and any $\theta \in (\theta_{i-1}, \theta_i)$, by definition of $D_i(\theta)$, $D_i'(\theta) = 2(\hat{\theta}_{i+1} - \hat{\theta}_i) - \frac{2q_i^2\{[q_i(1-2\Delta_i)+2\Delta_i](\theta-\hat{\theta}_i) + b[q_i(1-\Delta_i)+\Delta_i]\}}{[q_i(1-\Delta_i)+\Delta_i]^2}$ and $D_i''(\theta) = -\frac{2q_i^2[q_i(1-2\Delta_i)+2\Delta_i]}{[q_i(1-\Delta_i)+\Delta_i]^2} \le 0$ (because $q_i \in [0,1]$ and $\Delta_i \in (0,1)$).

Thus, by continuity of $D_i'(\theta)$ on (θ_{i-1}, θ_i), the infimum of $D_i'(\theta)$ on (θ_{i-1}, θ_i) is

$$
\lim_{\theta\uparrow\theta_i} D_i'(\theta) = 2(\hat{\theta}_{i+1} - \hat{\theta}_i) - \frac{2q_i^2\{[q_i(1-2\Delta_i)+2\Delta_i]\frac{\Delta_i}{2} + b[q_i(1-\Delta_i)+\Delta_i]\}}{[q_i(1-\Delta_i)+\Delta_i]^2}.
$$

For now, denote q_i as r and Δ_i as δ'. Let $J(r) := -\frac{r^2\{\delta'(\delta'+b) + r[(\delta'+b)(1-\delta') - \frac{\delta'}{2}]\}}{[r(1-\delta')+\delta']^2} + \hat{\theta}_{i+1} - \hat{\theta}_i$ for all $r \in [0,1]$. Accordingly, $\lim_{\theta\downarrow\theta_i} D_i'(\theta) = 2J(r)$, $J(0) = \hat{\theta}_{i+1} - \hat{\theta}_i > 0$, and $J(1) = \frac{\Delta_{i+1}}{2} - b \ge 0$ (by Claim 1 in the proof of Proposition 1).

Accordingly, $J'(r) = -\frac{r\{r[r(1-\delta')+3\delta'][(\delta'+b)(1-\delta') - \frac{\delta'}{2}] + 2\delta'^2(\delta'+b)\}}{[r(1-\delta')+\delta']^3}$, where $J'(0) = 0$, $J'(1) = -b - b\delta' - \frac{\delta'}{2} < 0$.

Next, let $K(r) := -r[r(1-\delta')+3\delta'][(\delta'+b)(1-\delta') - \frac{\delta'}{2}] - 2\delta'^2(\delta'+b)$ for all $r \in [0,1]$. Then, $K(1) = -b - b\delta' - \frac{\delta'}{2} < 0$, $K(0) = -2\delta'^2(\delta'+b) < 0$, $J(r) = \frac{rK(r)}{[r(1-\delta')+\delta']^3}$, and $K'(r) = -[(\delta'+b)(1-\delta') - \frac{\delta'}{2}][2r(1-\delta') + 3\delta']$.

Then, if $(\delta'+b)(1-\delta')-\frac{\delta'}{2}\geq 0$, then $K'(r)\leq 0$ on $[0,1]$. Thus, $K(r)\leq K(0)<0$ on $[0,1]$, implying $J'(r)\leq 0$ on $[0,1]$. Therefore, $J(r)\geq J(1)\geq 0$ on $[0,1]$.

Also, if $(\delta'+b)(1-\delta')-\frac{\delta'}{2}<0$, then $K'(r)>0$ on $[0,1]$. Thus, $K(r)\leq K(1)<0$ on $[0,1]$, implying $J'(r)\leq 0$ on $[0,1]$. Hence, $J(r)\geq J(1)\geq 0$ on $[0,1]$.

Therefore, in any case, for any $\theta\in(\theta_{i-1},\theta_i)$, $D_i'(\theta)\geq \lim_{\theta\uparrow\theta_i} D_i'(\theta)=2J(r)\geq 0$.

Then, since $D_i(\theta)$ is continuous on (θ_{i-1},θ_i), $D_i(\theta_i)=\lim_{\theta\uparrow\theta_i} D_i(\theta)$ and, when $i=1$, $D_i(\theta_{i-1})=\lim_{\theta\downarrow\theta_{i-1}} D_i(\theta)$, $\lim_{\theta\uparrow\theta_i} D_i(\theta)=\sup_{\theta\in\Theta_i} D_i(\theta)$.

Hence, $D_i(\theta)\leq 0$ for all $\theta\in\Theta_i$ if and only if $D_i^-|_\Delta=\lim_{\theta\uparrow\theta_i} D_i(\theta)\leq 0$. □

If.

First, for any $i\in I$ and any $\theta'\in\Theta_i$, by definition of $D_i(\theta)$, SR_m is satisfied for type θ' if, for any $j\in I$, $j\neq i$, $\begin{cases}\sum_{k=i}^{j-1} D_i(\theta')\leq 0 & \text{if } j>i\\ \sum_{k=j}^{i-1} D_i(\theta')\geq 0 & \text{if } j<i\end{cases}$, which is satisfied if for any $l\in I$, $D_l(\theta')\geq 0$ if $l<i$, and $D_l(\theta')\leq 0$ if $l\geq i$. Thus, it suffices to show that for any $i,j\in I$ and any $\theta\in\Theta_j$, $D_i(\theta)\geq 0$ if $j>i$, and $D_i(\theta)\leq 0$ if $j\leq i$.

Note that from the previous part of the proof, for any $i\in I\setminus\{n\}$, $D_i^+|_\Delta\geq 0\geq D_i^-|_\Delta$ implies that $D_i(\theta)\geq 0$ for any $\theta\in\Theta_{i+1}$ and $D_i(\theta)\leq 0$ for any $\theta\in\Theta_i$.

Next, for any $i\in I\setminus\{n\}$ and any $\theta\in\Theta\setminus(\Theta_i\cup\Theta_{i+1})$, by definition of $D_i(\theta)$, $D_i(\theta)=-(\hat\theta_{i+1}-\theta-b)^2-\frac{(1-q_{i+1})q_{i+1}^2\Delta_{i+1}^3}{12[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]^2}+(\hat\theta_i-\theta-b)^2+\frac{(1-q_i)q_i^2\Delta_i^3}{12[q_i(1-\Delta_i)+\Delta_i]^2}$, and for any $\theta\in(0,\theta_{i-1})\cup(\theta_{i+1},1)$, $D_i'(\theta)=\theta_{i+1}-\theta_{i-1}>0$.

Thus, for any $i,j\in I$, $j\neq i$, and any $\theta\in\Theta_j$, since $D_i(\theta_{j-1})=\lim_{\theta\downarrow\theta_{j-1}} D_i(\theta)$ when $\theta_{j-1}\in\Theta_j$ and $D_i(\theta_j)=\lim_{\theta\uparrow\theta_j} D_i(\theta)$ when $\theta_j\in\Theta_j$, $D_i(\theta)\geq 0$ for all $j\geq i+2$ if $\lim_{\theta\downarrow\theta_{i+1}} D_i(\theta)\geq 0$, and $D_i(\theta)\leq 0$ for all $j\leq i-1$ if $\lim_{\theta\uparrow\theta_{i-1}} D_i(\theta)\leq 0$.

Again by definition of $D_i(\theta)$, let

$$\begin{aligned}D_i^{++}|_\Delta &:= \lim_{\theta\downarrow\theta_{i+1}} D_i(\theta)=-\frac{(1-q_{i+1})q_{i+1}^2\Delta_{i+1}^3}{12[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]^2}+\frac{(1-q_i)q_i^2\Delta_i^3}{12[q_i(1-\Delta_i)+\Delta_i]^2}\\ &\quad+(\Delta_i+\Delta_{i+1})\left(\frac{3\Delta_{i+1}}{4}+\frac{\Delta_i}{4}+b\right),\\ D_i^{--}|_\Delta &:= \lim_{\theta\uparrow\theta_{i-1}} D_i(\theta)=-\frac{(1-q_{i+1})q_{i+1}^2\Delta_{i+1}^3}{12[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]^2}+\frac{(1-q_i)q_i^2\Delta_i^3}{12[q_i(1-\Delta_i)+\Delta_i]^2}\\ &\quad-(\Delta_i+\Delta_{i+1})\left(\frac{3\Delta_i}{4}+\frac{\Delta_{i+1}}{4}-b\right).\end{aligned}$$

Note that by definition of $D_i^-|_\Delta$,

$$D_i^-|_\Delta-D_i^{--}|_\Delta=\Delta_i\left\{(\Delta_i+\Delta_{i+1})+\frac{\{[\frac{1}{4}\Delta_i-(\frac{\Delta_i}{2}+b)(1-\Delta_i)]q_i^3-\Delta_i(\frac{\Delta_i}{2}+b)q_i^2\}}{[q_i(1-\Delta_i)+\Delta_i]^2}\right\}.$$

Then, for function $\psi(q):=\frac{[\frac{1}{4}\Delta_i-(\frac{\Delta_i}{2}+b)(1-\Delta_i)]q^3-\Delta_i(\frac{\Delta_i}{2}+b)q^2}{[q(1-\Delta_i)+\Delta_i]^2}$ defined on $[0,1]$ with $\Delta_i\in(0,1]$,

$\psi'(q) = \frac{q^2[(1-\Delta_i)q+3\Delta_i][\frac{1}{4}\Delta_i-(\frac{\Delta_i}{2}+b)(1-\Delta_i)]-2\Delta_i^2(\frac{\Delta_i}{2}+b)q}{[q(1-\Delta_i)+\Delta_i]^3}$.

Next, consider function $\chi(q) := q[(1-\Delta_i)q+3\Delta_i][\frac{1}{4}\Delta_i-(\frac{\Delta_i}{2}+b)(1-\Delta_i)]-2\Delta_i^2(\frac{\Delta_i}{2}+b)$ defined on $[0,1]$. Note that $\psi'(q) = \frac{\Delta_i\chi(q)}{[q(1-\Delta_i)+\Delta_i]^3}$.

With $\chi'(q) = [2(1-\Delta_i)q+3\Delta_i][\frac{1}{4}\Delta_i-(\frac{\Delta_i}{2}+b)(1-\Delta_i)]$, $\chi(0) = -2\Delta_i^2(\frac{\Delta_i}{2}+b) < 0$, and $\chi(1) = -\frac{\Delta_i}{4} - b(1+\Delta_i) < 0$, when $\frac{1}{4}\Delta_i-(\frac{\Delta_i}{2}+b)(1-\Delta_i) \geq 0$, $\chi(q) \leq \chi(1) < 0$ on $[0,1]$, and when $\frac{1}{4}\Delta_i-(\frac{\Delta_i}{2}+b)(1-\Delta_i) < 0$, $\chi(q) \leq \chi(0) < 0$ on $[0,1]$.

Thus, $\chi(q)$ is always negative on $[0,1]$, implying $\psi'(q) = \frac{\Delta_i\chi(q)}{[q(1-\Delta_i)+\Delta_i]^3} \leq 0$ on $[0,1]$. Therefore, for any $q \in [0,1]$, $\psi(q) \geq \psi(1) = -\frac{\Delta_i}{4} - b$.

Accordingly, since $\Delta_{i+1} \geq 2b > 0$ (by Claim 1 in the proof of Proposition 1), $D_i^-|_\Delta - D_i^{--}|_\Delta \geq \Delta_i[(\Delta_i+\Delta_{i+1}) - (0.25\Delta_i + b)] = \Delta_i(0.75\Delta_i + \Delta_{i+1} - b) > 0$.

Hence, $D_i^{--}|_\Delta = \lim_{\theta\uparrow\theta_{i-1}} D_i(\theta) < D_i^-|_\Delta \leq 0$.

Next, for function $\phi(q) := \frac{(1-q)q^2}{[q(1-\Delta'')+\Delta'']^2}$ defined on $[0,1]$ with $\Delta'' \in (0,1]$, $\phi'(q) = \frac{q[-(1-\Delta'')q^2-3\Delta''q+2\Delta'']}{[q(1-\Delta'')+\Delta'']^3}$, $\phi''(q) = \frac{2\Delta''[\Delta''-(2+\Delta'')q]}{[q(1-\Delta'')+\Delta'']^4}$. Then, with $\phi'(0) = 0 > \phi'(1) = -1$, $\phi''(\frac{\Delta''}{2+\Delta''}) = 0$, and $\phi''(q)[q - \frac{\Delta''}{2+\Delta''}] \leq 0$ on $[0,1]$, there exists $\tilde{q} \in (\frac{\Delta''}{2+\Delta''}, 1)$ so that $\phi'(\tilde{q}) = 0$, $\phi'(q) > 0$ on $(0,\tilde{q})$, and $\phi'(q) < 0$ on $(\tilde{q},1]$.

Hence, $\min_{q\in[0,1]} \phi(q) = \min\{\phi(0),\phi(1)\} = 0$, $\max_{q\in[0,1]} \phi(q) = \phi(\tilde{q})$.

Note that $\phi'(\tilde{q}) = 0$ implies $2\Delta''(1-\tilde{q}) = \tilde{q}[(1-\Delta'')\tilde{q}+\Delta'']$, yielding $\phi(\tilde{q}) = \frac{(1-\tilde{q})\tilde{q}^2}{[\tilde{q}(1-\Delta'')+\Delta'']^2} = \frac{\tilde{q}^3}{2\Delta''[\tilde{q}(1-\Delta'')+\Delta'']}$, which is increasing in $\tilde{q}$ on $[0,1]$ given $\Delta'' \in (0,1]$.

Thus, for any $\tilde{q} \in [0,1]$, $\phi(\tilde{q}) \leq \phi(1) = \frac{1}{2\Delta''}$, and therefore, with $\Delta_i, \Delta_{i+1} \in (0,1]$,

$$\begin{aligned} D_i^{++}|_\Delta &\geq \frac{\Delta_i^3}{12} \times 0 - \frac{\Delta_{i+1}^3}{12}\frac{1}{2\Delta_{i+1}} + (\Delta_i+\Delta_{i+1})(\frac{3\Delta_{i+1}}{4} + \frac{\Delta_i}{4} + b) \\ &= (\Delta_i+\Delta_{i+1})(\frac{\Delta_i}{4}+b) + \frac{3\Delta_i\Delta_{i+1}}{4} + \frac{3\Delta_{i+1}^2}{4} - \frac{\Delta_{i+1}^2}{24} > 0. \end{aligned}$$

□

A.3 Proof of Lemma 1

First, since $n \in \mathbb{N}_+ \setminus \{1\}$ and $b \in (0, \frac{1}{2n(n-1)})$, $\Delta^{CS}(n,b)$ is well defined.

Also, by definition of $\Delta^{CS}(n,b)$, $\Delta^{CS}(n,b) \in (0,1]^n$, $\sum_{i\in I} \Delta_i^{CS}(n,b) = 1$, and $\Delta_{i+1}^{CS}(n,b) = \Delta_i^{CS}(n,b) + 4b > 4b$ for all $i \in I \setminus \{n\}$, where $I = \{1,\ldots,n\}$.

Next, let $q_i := q^0[c,\bar{q},\Delta_i^{CS}(n,b)]$ for all $i \in I$. Then, by definition of $\mathcal{E}_n(b,c,\bar{q})$, it remains to show that $D_i^+|_{\Delta^{CS}(n,b)} \geq 0 \geq D_i^-|_{\Delta^{CS}(n,b)}$ for all $i \in I \setminus \{n\}$.

Note that, since $\Delta^{CS}_{i+1}(n,b) = \Delta^{CS}_i(n,b) + 4b$, for any $i \in I \setminus \{n\}$,

$$
\begin{aligned}
D^+_i|_{\Delta^{CS}(n,b)} &= \frac{(1-q_i)q_i^2\Delta^{CS}_i(n,b)^3}{12\{q_i[1-\Delta^{CS}_i(n,b)]+\Delta^{CS}_i(n,b)\}^2} + \frac{q_{i+1}^2\Delta^{CS}_{i+1}(n,b)[\frac{\Delta^{CS}_{i+1}(n,b)}{4}-b]}{q_{i+1}[1-\Delta^{CS}_{i+1}(n,b)]+\Delta^{CS}_{i+1}(n,b)} \\
&\quad + \frac{(1-q_{i+1})q_{i+1}^2\Delta^{CS}_{i+1}(n,b)^3}{6\{q_{i+1}[1-\Delta^{CS}_{i+1}(n,b)]+\Delta^{CS}_{i+1}(n,b)\}^2}, \\
D^-_i|_{\Delta^{CS}(n,b)} &= -\frac{(1-q_{i+1})q_{i+1}^2\Delta^{CS}_{i+1}(n,b)^3}{12\{q_{i+1}[1-\Delta^{CS}_{i+1}(n,b)]+\Delta^{CS}_{i+1}(n,b)\}^2} \\
&\quad - \frac{q_i^2\Delta^{CS}_i(n,b)[\frac{\Delta^{CS}_i(n,b)}{4}+b]}{q_i[1-\Delta^{CS}_i(n,b)]+\Delta^{CS}_i(n,b)} - \frac{(1-q_i)q_i^2\Delta^{CS}_i(n,b)^3}{6\{q_i[1-\Delta^{CS}_i(n,b)]+\Delta^{CS}_i(n,b)\}^2},
\end{aligned}
$$

while $\min\left\{\frac{q_i^2\Delta^{CS}_i(n,b)[\frac{\Delta^{CS}_i(n,b)}{4}+b]}{q_i[1-\Delta^{CS}_i(n,b)]+\Delta^{CS}_i(n,b)}, \frac{(1-q_i)q_i^2\Delta^{CS}_i(n,b)^3}{\{q_i[1-\Delta^{CS}_i(n,b)]+\Delta^{CS}_i(n,b)\}^2}, \frac{q_{i+1}^2\Delta^{CS}_{i+1}(n,b)[\frac{\Delta^{CS}_{i+1}(n,b)}{4}-b]}{q_{i+1}[1-\Delta^{CS}_{i+1}(n,b)]+\Delta^{CS}_{i+1}(n,b)}, \frac{(1-q_{i+1})q_{i+1}^2\Delta^{CS}_{i+1}(n,b)^3}{\{q_{i+1}[1-\Delta^{CS}_{i+1}(n,b)]+\Delta^{CS}_{i+1}(n,b)\}^2}\right\} \geq 0$ because $q_i, q_{i+1} \in [0,1]$, $\Delta^{CS}_i(n,b) > 0$, and $\Delta^{CS}_{i+1}(n,b) > 4b$.

Accordingly, $D^+_i|_{\Delta^{CS}(n,b)} \geq 0 \geq D^-_i|_{\Delta^{CS}(n,b)}$ for all $i \in I \setminus \{n\}$. □

A.4 Proof of Corollary 1

First, since $n \in \mathbb{N}_+ \setminus \{1\}$ and $b \in (0, \frac{1}{2n(n-1)})$, $\Delta^{CS}(n,b)$ is well defined.

Second, since $c < \frac{\Delta^{CS}_n(n,b)}{12}$ and $\bar{q} > 0$, by SR'_q, $q_n := q^0(c, \bar{q}, \Delta_n) > 0$.

Also, by definition of $\Delta^{CS}(n,b)$, $\Delta^{CS}_n(n,b) = \Delta^{CS}_{n-1}(n,b) + 4b > 4b$.

Now, define a partition $\Delta(\epsilon) = (\Delta_1(\epsilon), \ldots, \Delta_n(\epsilon))$ for some $\epsilon \in (0, \Delta^{CS}_1(n,b))$ so that $\Delta_n(\epsilon) = \Delta^{CS}_n(n,b) + (n-1)\epsilon$ and $\Delta_i(\epsilon) = \Delta^{CS}_i(n,b) - \epsilon$ for all $i \in \{1, \ldots, n-1\} = I \setminus \{n\}$. Note that, by definition of $\Delta(\epsilon)$, $\sum_{i \in I} \Delta_i(\epsilon) = 1$, $\Delta(\epsilon) \in (0,1)^n$, and for any $i \in I \setminus \{n, n-1\}$, $\Delta_{i+1}(\epsilon) = \Delta_i(\epsilon) + 4b$.

Note that $\Delta^{CS}_n(n,b) = \Delta^{CS}_{n-1}(n,b) + 4b$ implies $[0.5\Delta^{CS}_{n-1}(n,b) + b]^2 - [0.5\Delta^{CS}_n(n,b) - b]^2 = 0$. Then, since $n \in \mathbb{N}_+ \setminus \{1\}$ and $\Delta^{CS}_n(n,b) > 4b$,

$$
\begin{aligned}
D_{n-1}|_{\Delta(\epsilon)} &:= -[0.5\Delta_n(\epsilon) - b]^2 + [0.5\Delta_{n-1}(\epsilon) + b]^2 \\
&= -\{0.5[\Delta^{CS}_n(n,b) + (n-1)\epsilon] - b\}^2 + \{0.5[\Delta^{CS}_{n-1}(n,b) - \epsilon] + b\}^2 \\
&= 0 - (n-1)\epsilon[0.5\Delta^{CS}_n(n,b) - b] - \epsilon[0.5\Delta^{CS}_{n-1}(n,b) + b] \\
&\quad - 0.25(n-1)^2\epsilon^2 - 0.25\epsilon^2 < 0.
\end{aligned}
$$

Next, let $q_i(\epsilon) := q^0[c, \bar{q}, \Delta_i(\epsilon)]$ for all $i \in I$. Then, following the argument in the proof of Lemma 1, for any $i \in I \setminus \{n, n-1\}$, since $\Delta_{i+1}(\epsilon) = \Delta_i(\epsilon) + 4b$ and $q_i(\epsilon), q_{i+1}(\epsilon) \in [0,1]$, by definition of D^+_i and D^-_i, $D^+_i|_{\Delta(\epsilon)} \geq 0 \geq D^-_i|_{\Delta(\epsilon)}$.

Similarly, since $D_{n-1}|_{\Delta(\epsilon)} < 0$ and $q_{n-1}(\epsilon), q_n(\epsilon) \in [0,1]$, $D^-_{n-1}|_{\Delta(\epsilon)} < 0$.

Finally, note that $q_n > 0$ and $\Delta_n^{CS}(n,b) > 4b$ imply $D_{n-1}^+|_{\Delta^{CS}(n,b)} > 0$. Thus, by definition of $\Delta_n(\epsilon)$ and $\Delta_{n-1}(\epsilon)$, since $D_{n-1}^+|_\Delta$ is continuous in Δ on $(0,1]^n$, there exists some small $\hat{\epsilon} > 0$ so that $D_{n-1}^+|_{\Delta(\epsilon)} > 0$ for all $\epsilon \in (0, \hat{\epsilon})$.

Accordingly, let $\bar{\epsilon} := \min\{\hat{\epsilon}, \Delta_1^{CS}(n,b)\}$, then $\bar{\epsilon} > 0$ and $\Delta(\epsilon) \in \mathcal{E}_n(b,c,\bar{q})$ for all $\epsilon \in (0,\bar{\epsilon})$. Hence, $\mathcal{E}_n(b,c,\bar{q})$ contains a continuum of partitions $\{\Delta(\epsilon)\}_{\epsilon\in(0,\bar{\epsilon})}$ and is therefore uncountable because the set $(0,\bar{\epsilon})$ is uncountable. □

A.5 Proof of Lemma 2

First, since $\bar{W}^\alpha(b,c,\bar{q}) = \max_{n\in\mathcal{N}(b,c,\bar{q})} \max_{\Delta\in\bar{\mathcal{E}}_n(b,c,\bar{q})} W^\alpha(b,c,\bar{q})|_\Delta$, there exists some $n' \in \mathcal{N}(b,c,\bar{q})$ and some $\Delta' \in \bar{\mathcal{E}}_{n'}(b,c,\bar{q})$ so that $W^\alpha(b,c,\bar{q})|_{\Delta'} = \bar{W}^\alpha(b,c,\bar{q})$.

If $\Delta' \in \mathcal{E}_{n'}(b,c,\bar{q})$, then let $n = n'$, $\Delta = \Delta'$, the lemma is proved.

If $\Delta' \in \bar{\mathcal{E}}_{n'}(b,c,\bar{q}) \setminus \mathcal{E}_{n'}(b,c,\bar{q})$, since $\mathcal{E}_1(b,c,\bar{q}) = \bar{\mathcal{E}}_1(b,c,\bar{q}) = \{(1)\}$, $\Delta' \neq (1)$ and $n' \geq 2$. Also, there must exist some $i' \in I'$ so that $\Delta'_{i'} = 0$, where $I' := \{1,\dots,n'\}$ is the associated index set of Δ'. Let $I'_0 := \{i' \in I' : \Delta'_{i'} = 0\}$, $n := n' - \#I'_0 \in \mathbb{N}_+$, and $q'_{i'} := q^0(c,\bar{q},\Delta'_{i'})$ for all $i' \in I'$. Then, $q'_{i'} = 0$ for all $i' \in I'_0$.

Next, define $\Delta \in [0,1]^n$ so that $\Delta_{i'-i_0^{i'}} = \Delta'_{i'}$ for all $i' \in I' \setminus I'_0$, where $i_0^{i'} := \#I_0^{i'}$ and $I_0^{i'} := \{j \in I'_0 : j \leq i'\}$ for all $i' \in \mathbb{N}_+$. Note that $\{i \in \mathbb{N}_+ : i = i' - i_0^{i'}, i' \in I' \setminus I'_0\} = \{1,\dots,n\} =: I$ and, by definition of I'_0 and Δ, $\Delta \in (0,1]^n$. Also, let $q_i := q^0(c,\bar{q},\Delta_i)$ for all $i \in I$, then $q_{i'-i_0^{i'}} = q'_{i'}$ for all $i' \in I' \setminus I'_0$.

Note that, if $n = 1$, then $\Delta = (1) \in \mathcal{E}_1(b,c,\bar{q})$. Also, if $n \geq 2$, by definition of Δ, for any $i \in I \setminus \{n\}$, there exists $i' \in I'$ so that $\Delta_i = \Delta'_{i'}$ and either (1) $\Delta_{i+1} = \Delta'_{i'+1}$ or (2) there exists $j \in \mathbb{N}_+ \setminus \{1\}$ so that $\Delta_{i+1} = \Delta'_{i'+j}$ and $\Delta'_{i'+k} = 0$ for all $k \in \mathbb{N}_+$, $k < j$. Then, with $\Delta' \in \bar{\mathcal{E}}_{n'}(b,c,\bar{q})$, when $n \geq 2$, for any $i \in I \setminus \{n\}$:

1. if there exists $i' \in I'$ so that $\Delta_i = \Delta'_{i'}$ and $\Delta_{i+1} = \Delta'_{i'+1}$, $D_i^+|_\Delta = D_{i'}^+|_{\Delta'} \geq 0 \geq D_{i'}^-|_{\Delta'} = D_i^-|_\Delta$;

2. if there exists $i' \in I'$ and $j \in \mathbb{N}_+$, $j \geq 2$ so that $\Delta_i = \Delta'_{i'}$, $\Delta_{i+1} = \Delta'_{i'+j}$ and $\Delta'_{i'+k} = 0$ for all $k \in \mathbb{N}_+$, $k < j$, since $D_k^+|_{\Delta'} \geq 0 \geq D_k^-|_{\Delta'}$ for all $k \in \{i', i'+1, \dots, i'+j-1\}$, $D_i^+|_\Delta = \sum_{k=i'}^{i'+j-1} D_k^+|_{\Delta'} \geq 0 \geq \sum_{k=i'}^{i'+j-1} D_k^-|_{\Delta'} = D_i^-|_\Delta$.

Thus, in any case, $\Delta \in \mathcal{E}_n(b,c,\bar{q})$. Hence, $\mathcal{E}_n(b,c,\bar{q}) \neq \varnothing$, indicating $n \in \mathcal{N}(b,c,\bar{q})$. Also,

since $\bar{W}^\alpha(b,c,\bar{q}) = W^\alpha(b,c,\bar{q})|_{\Delta'}$, by definition of Δ and I'_0,

$$\begin{aligned}\bar{W}^\alpha(b,c,\bar{q}) &= -\alpha b^2 - (1-\alpha)c\sum_{i\in I'}\Delta'_i q_i'^2 - \sum_{i\in I'}\mathbb{1}[\Delta'_i>0]\frac{\Delta_i'^3(1-q'_i)(\Delta'_i+q'_i)}{12[q'_i(1-\Delta'_i)+\Delta'_i]}\\ &= -\alpha b^2 - (1-\alpha)c\sum_{i\in I'\setminus I'_0}\Delta'_i q_i'^2 - \sum_{i\in I'\setminus I'_0}\frac{\Delta_i'^3(1-q'_i)(\Delta'_i+q'_i)}{12[q'_i(1-\Delta'_i)+\Delta'_i]}\\ &= -\alpha b^2 - (1-\alpha)c\sum_{i\in I}\Delta_i q_i^2 - \sum_{i\in I}\frac{\Delta_i^3(1-q_i)(\Delta_i+q_i)}{12[q_i(1-\Delta_i)+\Delta_i]} = W^\alpha(b,c,\bar{q})|_\Delta.\end{aligned}$$

□

A.6 Proof of Proposition 3

First, when $\Delta = (1) \in (0,1]$, $\frac{\Delta_M}{12} = \frac{\Delta_M^2[\bar{q}(1-\Delta_M)+2\Delta_M]}{24[\bar{q}(1-\Delta_M)+\Delta_M]^2} = \frac{1}{12}$ and, by SR'_q,

$$W^\alpha(b,c,\bar{q})|_\Delta = \begin{cases} -\alpha b^2 - \frac{1}{12} & \text{if } c \geq \frac{1}{12} \\ -\alpha b^2 - (1-\alpha)c\bar{q}^2 - \frac{(1-\bar{q})(1+\bar{q})}{12} & \text{if } c < \frac{1}{12}\end{cases}.$$

Accordingly,

$$W^\alpha(b,c',\bar{q})|_\Delta - W^\alpha(b,c,\bar{q})|_\Delta = \begin{cases} 0 & \text{if } c' \geq \frac{1}{12} \\ -(1-\alpha)c'\bar{q}^2 + \frac{\bar{q}^2}{12} & \text{if } c \geq \frac{1}{12} > c' \\ -(1-\alpha)(c'-c)\bar{q}^2 & \text{if } c < \frac{1}{12}\end{cases}$$

$$\geq 0 \text{ (inequality is strict if and only if } \bar{q}>0,\ c' < \frac{1}{12},$$

$$\text{and, when } c < \frac{1}{12},\ \alpha < 1),$$

$$W^\alpha(b,c,\bar{q}')|_\Delta - W^\alpha(b,c,\bar{q})|_\Delta = \begin{cases} 0 & \text{if } c \geq \frac{1}{12} \\ [\frac{1}{12} - (1-\alpha)c](\bar{q}'^2 - \bar{q}^2) & \text{if } c < \frac{1}{12}\end{cases}$$

$$\geq 0 \text{ (inequality is strict if and only if } c < \frac{1}{12}).$$

Next, consider cases with partition $\Delta \in (0,1]^n$, $n \geq 2$.

For any $(b,c,\bar{q}) \in \mathbb{R}^2_{++}\times[0,1]$ and any $\alpha \in [0,1]$, rewrite $W^\alpha(b,c,\bar{q})|_\Delta = \sum_{i\in I} W_i^\alpha(b,c,\bar{q})|_\Delta$, where for any $i \in I = \{1,\ldots,n\}$, with $q_i := q^0(c,\bar{q},\Delta_i)$ and $\Delta_i \in (0,1]$,

$$W_i^\alpha(b,c,\bar{q})|_\Delta := -\alpha b^2\Delta_i - (1-\alpha)c\Delta_i q_i^2 - \frac{\Delta_i^3(1-q_i)(\Delta_i+q_i)}{12[q_i(1-\Delta_i)+\Delta_i]}.$$

Part 1.

For any $\hat{c}, \tilde{c} \in \mathbb{R}_{++}$ so that $\hat{c} > \tilde{c}$ and any $i \in I$, let $\hat{q}_i := q^0(\hat{c}, \bar{q}, \Delta_i)$ and $\tilde{q}_i := q^0(\tilde{c}, \bar{q}, \Delta_i)$. When $\tilde{c} \geq \frac{\Delta_i}{12}$, by SR'_q, $\hat{q}_i = \tilde{q}_i = 0$. Therefore, $W_i^\alpha(b, \hat{c}, \bar{q})|_\Delta = W_i^\alpha(b, \tilde{c}, \bar{q})|_\Delta = -\alpha b^2 \Delta_i - \frac{\Delta_i^3}{12}$, yielding $W_i^\alpha(b, \tilde{c}, \bar{q})|_\Delta - W_i^\alpha(b, \hat{c}, \bar{q})|_\Delta = 0$.

Also, when $\frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2} \leq \tilde{c} < \hat{c} \leq \frac{\Delta_i}{12}$ (requiring $\frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2} < \frac{\Delta_i}{12}$, i.e., $\bar{q} > 0$), by SR'_q, $\hat{c} = \frac{\Delta_i^2[\hat{q}_i(1-\Delta_i)+2\Delta_i]}{24[\hat{q}_i(1-\Delta_i)+\Delta_i]^2}$, $\tilde{c} = \frac{\Delta_i^2[\tilde{q}_i(1-\Delta_i)+2\Delta_i]}{24[\tilde{q}_i(1-\Delta_i)+\Delta_i]^2}$, and $\tilde{q}_i > \hat{q}_i$. Thus,

$$\begin{aligned}
\hat{c}\hat{q}_i^2 - \tilde{c}\tilde{q}_i^2 = \Delta_i^2 \times \{24[\hat{q}_i(1-\Delta_i)+\Delta_i]^2[\tilde{q}_i(1-\Delta_i)+\Delta_i]^2\}^{-1} \\
\times \{(\hat{q}_i - \tilde{q}_i)[(1-\Delta_i)^3\hat{q}_i^2\tilde{q}_i^2 + 4\Delta_i^2(1-\Delta_i)\hat{q}_i\tilde{q}_i] + 2\Delta_i^3(\hat{q}_i^2 - \tilde{q}_i^2) \\
+ 2\Delta_i(1-\Delta_i)^2\hat{q}_i\tilde{q}_i(\hat{q}_i^2 - \tilde{q}_i^2) + \Delta_i^2(1-\Delta_i)(\hat{q}_i^3 - \tilde{q}_i^3)\} < 0.
\end{aligned}$$

Then, combined with $\alpha \in [0,1]$, for $\Delta_c W_i^\alpha := W_i^\alpha(b, \tilde{c}, \bar{q})|_\Delta - W_i^\alpha(b, \hat{c}, \bar{q})|_\Delta$,

$$\begin{aligned}
\Delta_c W_i^\alpha &= (1-\alpha)\Delta_i(\hat{c}\hat{q}_i^2 - \tilde{c}\tilde{q}_i^2) - \frac{\Delta_i^3(1-\tilde{q}_i)(\Delta_i+\tilde{q}_i)}{12[\tilde{q}_i(1-\Delta_i)+\Delta_i]} + \frac{\Delta_i^3(1-\hat{q}_i)(\Delta_i+\hat{q}_i)}{12[\hat{q}_i(1-\Delta_i)+\Delta_i]} \\
&\geq -\tilde{c}\Delta_i\tilde{q}_i^2 + \frac{\Delta_i^3\tilde{q}_i^2}{12[\tilde{q}_i(1-\Delta_i)+\Delta_i]} + \hat{c}\Delta_i\hat{q}_i^2 - \frac{\Delta_i^3\hat{q}_i^2}{12[\hat{q}_i(1-\Delta_i)+\Delta_i]} \\
&= \frac{\Delta_i^3}{24}\left\{\frac{(1-\Delta_i)\tilde{q}_i^3}{[\tilde{q}_i(1-\Delta_i)+\Delta_i]^2} - \frac{(1-\Delta_i)\hat{q}_i^3}{[\hat{q}_i(1-\Delta_i)+\Delta_i]^2}\right\} > 0.
\end{aligned}$$

Finally, when $\hat{c} \leq \frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}$, $\hat{q}_i = \tilde{q}_i = \bar{q}$. Thus,

$$W_i^\alpha(b, \tilde{c}, \bar{q})|_\Delta - W_i^\alpha(b, \hat{c}, \bar{q})|_\Delta = -(1-\alpha)(\tilde{c} - \hat{c})\Delta_i\bar{q}^2 \geq 0,$$

where the inequality is strict if and only if $\bar{q} > 0$ and $\alpha < 1$.

Thus, in any case, for any $i \in I$, $W_i^\alpha(b, c', \bar{q})|_\Delta - W_i^\alpha(b, c, \bar{q})|_\Delta \geq 0$, and the inequality is strict if and only if $\bar{q} > 0$, $c' < \frac{\Delta_i}{12}$, and, when $c \leq \frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}$, $\alpha < 1$.

Hence, in any case, $W^\alpha(b, c', \bar{q})|_\Delta - W^\alpha(b, c, \bar{q})|_\Delta \geq 0$, and the inequality is strict if and only if $\bar{q} > 0$, $c' < \max\left\{\frac{\Delta_i}{12}\right\}_{i\in I} = \frac{\Delta_M}{12}$, and, when $c \leq \min\left\{\frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}\right\}_{i\in I} = \frac{\Delta_m^2[\bar{q}(1-\Delta_m)+2\Delta_m]}{24[\bar{q}(1-\Delta_m)+\Delta_m]^2}$, $\alpha < 1$. □

Part 2.

First, when $c \geq \frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}$, by SR'_q, $q^0(c, \bar{q}, \Delta_i) = q^0(c, \bar{q}', \Delta_i)$. Therefore, $W_i^\alpha(b, c, \bar{q})|_\Delta = W_i^\alpha(b, c, \bar{q}')|_\Delta$, i.e., $W_i^\alpha(b, c, \bar{q}')|_\Delta - W_i^\alpha(b, c, \bar{q})|_\Delta = 0$.

Second, when $c < \frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2} \leq \frac{\Delta_i}{12}$, by SR'_q, $q_i := q^0(c, \bar{q}, \Delta_i) = \bar{q} < q^0(c, \bar{q}', \Delta_i) =: q'_i$.

Thus, since $\alpha \in [0,1]$, for $\Delta_{\bar{q}} W_i^\alpha := W_i^\alpha(b,c,\bar{q}')|_\Delta - W_i^\alpha(b,c,\bar{q})|_\Delta$,

$$\begin{aligned}
\Delta_{\bar{q}} W_i^\alpha &= (1-\alpha)c\Delta_i(q_i^2 - q_i'^2) + \frac{\Delta_i^3 q_i'^2}{12[q_i'(1-\Delta_i)+\Delta_i]} - \frac{\Delta_i^3 q_i^2}{12[q_i(1-\Delta_i)+\Delta_i]} \\
&\geq \frac{\Delta_i^2}{12}(q_i^2 - q_i'^2) + \frac{\Delta_i^3 q_i'^2}{12[q_i'(1-\Delta_i)+\Delta_i]} - \frac{\Delta_i^3 q_i^2}{12[q_i(1-\Delta_i)+\Delta_i]} \\
&= \frac{\Delta_i^2(1-\Delta_i)}{12}\Big[\frac{q_i'^3}{q_i'(1-\Delta_i)+\Delta_i} - \frac{q_i^3}{q_i(1-\Delta_i)+\Delta_i}\Big] > 0.
\end{aligned}$$

Therefore, in any case, for any $i \in I$, $W_i^\alpha(b,c,\bar{q}')|_\Delta - W_i^\alpha(b,c,\bar{q})|_\Delta \geq 0$, and the inequality is strict if and only if $c < \frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}$.

Accordingly, in any case, $W^\alpha(b,c,\bar{q}')|_\Delta - W^\alpha(b,c,\bar{q})|_\Delta \geq 0$, and the inequality is strict if and only if $c < \max\left\{\frac{\Delta_i^2[\bar{q}(1-\Delta_i)+2\Delta_i]}{24[\bar{q}(1-\Delta_i)+\Delta_i]^2}\right\}_{i\in I} = \frac{\Delta_M^2[\bar{q}(1-\Delta_M)+2\Delta_M]}{24[\bar{q}(1-\Delta_M)+\Delta_M]^2}$. □

A.7 Proof of Corollary 2

Since $\Delta \in \mathcal{E}_n(b,c',\bar{q}) \subset \bar{\mathcal{E}}_n(b,c',\bar{q})$, which also implies that $n \in \mathcal{N}(b,c',\bar{q})$, by part 1 of Proposition 3,

$$\begin{aligned}
\bar{W}^\alpha(b,c,\bar{q}) &= W^\alpha(b,c,\bar{q})|_\Delta \\
&\leq W^\alpha(b,c',\bar{q})|_\Delta \text{ (inequality is strict if } \bar{q}>0,\ c' < \frac{\Delta_M}{12}, \text{ and } \alpha < 1) \\
&\leq \max_{\Delta \in \bar{\mathcal{E}}_n(b,c',\bar{q})} W^\alpha(b,c',\bar{q})|_\Delta \\
&\leq \max_{n \in \mathcal{N}(b,c',\bar{q})} \max_{\Delta \in \bar{\mathcal{E}}_n(b,c',\bar{q})} W^\alpha(b,c',\bar{q})|_\Delta = \bar{W}^\alpha(b,c',\bar{q}).
\end{aligned}$$

Similarly, since $\Delta \in \mathcal{E}_n(b,c,\bar{q}') \subset \bar{\mathcal{E}}_n(b,c,\bar{q}')$, which also implies that $n \in \mathcal{N}(b,c,\bar{q}')$, by part 2 of Proposition 3,

$$\begin{aligned}
\bar{W}^\alpha(b,c,\bar{q}) &= W^\alpha(b,c,\bar{q})|_\Delta \\
&\leq W^\alpha(b,c,\bar{q}')|_\Delta \text{ (inequality is strict if } c < \frac{\Delta_M^2[\bar{q}(1-\Delta_M)+2\Delta_M]}{24[\bar{q}(1-\Delta_M)+\Delta_M]^2}) \\
&\leq \max_{\Delta \in \bar{\mathcal{E}}_n(b,c,\bar{q}')} W^\alpha(b,c,\bar{q}')|_\Delta \\
&\leq \max_{n \in \mathcal{N}(b,c,\bar{q}')} \max_{\Delta \in \bar{\mathcal{E}}_n(b,c,\bar{q}')} W^\alpha(b,c,\bar{q}')|_\Delta = \bar{W}^\alpha(b,c,\bar{q}').
\end{aligned}$$

□

A.8 Proof of Proposition 4

First, Crawford and Sobel (1982) showed that, in the CS model, both parties strictly prefer the equilibrium partition with the most intervals to the other equilibrium partitions.

Thus, by definition of $N^{CS}(b)$, $W^{\alpha}(b,c,0)|_{\Delta^{CS}[N^{CS}(b),b]} = \bar{W}^{\alpha}(b,c,0)$ and $\Delta_M^{CS}(b)$ is the length of the longest interval of the partition $\Delta^{CS}[N^{CS}(b),b]$.

Next, note that, by Lemma 1, $\Delta^{CS}[N^{CS}(b),b] \in \mathcal{E}_{N^{CS}(b)}(b,c,0) \bigcap \mathcal{E}_{N^{CS}(b)}(b,c,\bar{q}')$. Then, by part 2 of Corollary 2, $\bar{W}^{\alpha}(b,c,\bar{q}') - \bar{W}^{\alpha}(b,c,0) \geq 0$, and the inequality is strict if $c < \frac{\Delta_M^{CS}(b)^2\{0\times[1-\Delta_M^{CS}(b)]+2\Delta_M^{CS}(b)\}}{24\{0\times[1-\Delta_M^{CS}(b)]+\Delta_M^{CS}(b)\}^2} = \frac{\Delta_M^{CS}(b)}{12}$. □

A.9 Proof of Lemma 3

First, by Proposition 2, with $q_j := q^0(c,\bar{q},\Delta_j) \in [0,1]$ for all $j \in I := \{1,\ldots,n\}$ and under parameter values $(b,c,\bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$, for any $i \in I \setminus \{n\}$,

$$
\begin{aligned}
D_i^+|_{\Delta,(b,c,\bar{q})} &= -(\frac{\Delta_{i+1}}{2} - b)^2 + (\frac{\Delta_i}{2} + b)^2 + \frac{q_{i+1}^2\Delta_{i+1}(\frac{\Delta_{i+1}}{4} - b)}{q_{i+1}(1-\Delta_{i+1}) + \Delta_{i+1}} \\
&\quad + \frac{(1-q_{i+1})q_{i+1}^2\Delta_{i+1}^3}{6[q_{i+1}(1-\Delta_{i+1}) + \Delta_{i+1}]^2} + \frac{(1-q_i)q_i^2\Delta_i^3}{12[q_i(1-\Delta_i) + \Delta_i]^2}, \\
D_i^-|_{\Delta,(b,c,\bar{q})} &= -(\frac{\Delta_{i+1}}{2} - b)^2 + (\frac{\Delta_i}{2} + b)^2 - \frac{q_i^2\Delta_i(\frac{\Delta_i}{4} + b)}{q_i(1-\Delta_i) + \Delta_i} \\
&\quad - \frac{(1-q_{i+1})q_{i+1}^2\Delta_{i+1}^3}{12[q_{i+1}(1-\Delta_{i+1}) + \Delta_{i+1}]^2} - \frac{(1-q_i)q_i^2\Delta_i^3}{6[q_i(1-\Delta_i) + \Delta_i]^2}.
\end{aligned}
$$

Second, for any $i \in I \setminus \{n\}$, $\frac{\partial}{\partial q_i}\left[\frac{q_i^2\Delta_i(\frac{\Delta_i}{4}+b)}{q_i(1-\Delta_i)+\Delta_i}\right] = \frac{q_i[q_i(1-\Delta_i)+2\Delta_i]\Delta_i(\frac{\Delta_i}{4}+b)}{[q_i(1-\Delta_i)+\Delta_i]^2} \geq 0$.

Also, for any $i \in I\setminus\{n\}$, if $\Delta_{i+1} \leq 4b$, by SR'_q, $q_{i+1} = 0$ when $c \geq \frac{b}{3}$, implying $\frac{q_{i+1}^2\Delta_{i+1}(\frac{\Delta_{i+1}}{4}-b)}{q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}} = 0$, and if $\Delta_{i+1} > 4b$, $\frac{\partial}{\partial q_{i+1}}\left[\frac{q_{i+1}^2\Delta_{i+1}(\frac{\Delta_{i+1}}{4}-b)}{q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}}\right] = \frac{q_{i+1}[q_{i+1}(1-\Delta_{i+1})+2\Delta_{i+1}]\Delta_{i+1}(\frac{\Delta_{i+1}}{4}-b)}{[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]^2} \geq 0$.

Next, for any $i \in I$, if $\Delta_i \leq 4b$, by SR'_q, $q_i = 0$ when $c \geq \frac{b}{3}$, implying $\frac{(1-q_i)q_i^2}{[q_i(1-\Delta_i)+\Delta_i]^2} = 0$, and if $\Delta_i > 4b$, since $\frac{\partial}{\partial q_i}\left\{\frac{(1-q_i)q_i^2}{[q_i(1-\Delta_i)+\Delta_i]^2}\right\} = \frac{q_i[-q_i^2(1-\Delta_i)-3\Delta_i q_i+2\Delta_i]}{[q_i(1-\Delta_i)+\Delta_i]^3}$, $\frac{\partial}{\partial q_i}[-q_i^2(1-\Delta_i)-3\Delta_i q_i+2\Delta_i] = -2q_i(1-\Delta_i)-3\Delta_i < 0$, and $\frac{\partial}{\partial \Delta_i}[-q_i^2(1-\Delta_i)-3\Delta_i q_i+2\Delta_i] = (q_i-1)(q_i-2) \geq 0$ (because $\Delta_i \in (0,1]$ and $q_i \in [0,1]$), when $\bar{q} \leq \bar{\bar{q}}$ (which implies that $q_i \leq \bar{\bar{q}}$), $-q_i^2(1-\Delta_i)-3\Delta_i q_i+2\Delta_i \geq -\bar{\bar{q}}^2(1-4b)-12b\bar{\bar{q}}+8b = 0$, implying $\frac{\partial}{\partial q_i}\left\{\frac{(1-q_i)q_i^2}{[q_i(1-\Delta_i)+\Delta_i]^2}\right\} \geq 0$.

Finally, since q_j is non-increasing in c on $\mathbb{R}_{++}$ and non-decreasing in $\bar{q}$ on $[0,1]$ for all $j \in I$, with $c > c' \geq \frac{b}{3}$ and $0 \leq \bar{q} < \bar{q}' \leq \bar{\bar{q}}$, for any $i \in I\setminus\{n\}$ and any $Q \in \left\{\frac{q_{i+1}^2\Delta_{i+1}(\frac{\Delta_{i+1}}{4}-b)}{q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}}, \frac{(1-q_{i+1})q_{i+1}^2\Delta_{i+1}^3}{[q_{i+1}(1-\Delta_{i+1})+\Delta_{i+1}]^2},\right.$

$\frac{(1-q_i)q_i^2\Delta_i^3}{[q_i(1-\Delta_i)+\Delta_i]^2}, \frac{q_i^2\Delta_i(\frac{\Delta_i}{4}+b)}{q_i(1-\Delta_i)+\Delta_i}\Big\}$,

$$Q|_{(b,c,\bar{q})} \leq \min\left\{Q|_{(b,c',\bar{q})}, Q|_{(b,c,\bar{q}')}\right\}.$$

Hence, since $\Delta \in \mathcal{E}_n(b,c,\bar{q})$, $\min\left\{D_i^+|_{\Delta,(b,c',\bar{q})}, D_i^+|_{\Delta,(b,c,\bar{q}')}\right\} \geq D_i^+|_{\Delta,(b,c,\bar{q})} \geq 0 \geq D_i^-|_{\Delta,(b,c,\bar{q})}$ $\geq \max\left\{D_i^-|_{\Delta,(b,c',\bar{q})}, D_i^-|_{\Delta,(b,c,\bar{q}')}\right\}$ for all $i \in I \setminus \{n\}$, implying $\Delta \in \mathcal{E}_n(b,c',\bar{q})$ and $\Delta \in \mathcal{E}_n(b,c,\bar{q}')$. □

A.10 Proof of Proposition 5

First, by Lemma 2, there exists some $n \in \mathcal{N}(b,c,\bar{q})$ and some $\Delta \in \mathcal{E}_n(b,c,\bar{q})$ so that $W^\alpha(b,c,\bar{q})|_\Delta = \bar{W}^\alpha(b,c,\bar{q})$.

Second, by Lemma 3, since $0 \leq \bar{q} \leq \bar{\bar{q}}$ and $c > c' \geq \frac{b}{3}$, $\Delta \in \mathcal{E}_n(b,c',\bar{q})$.

Third, since $\bar{q} > 0$, by part 1 of Corollary 2, $\bar{W}^\alpha(b,c',\bar{q}) - \bar{W}^\alpha(b,c,\bar{q}) \geq 0$, and the inequality is strict if $c' < \frac{\Delta_M}{12}$ and $\alpha < 1$, where $\Delta_M = \max\{\Delta_i\}_{i\in I}$.

Next, the following lemma provides a sufficient condition for $c' < \frac{\Delta_M}{12}$.

Lemma 4. *For any $(b,c,\bar{q}) \in \mathbb{R}^2_{++} \times [0,1]$, any $n \in \mathbb{N}_+$, and any partition $\Delta \in \mathcal{E}_n(b,c,\bar{q})$, if $c < \frac{\Delta_M^{CS}(b)}{12}$, then $c < \frac{\Delta_M}{12}$.*

Proof of Lemma 4

Suppose not, i.e., $c \geq \frac{\Delta_i}{12}$ for all $i \in I = \{1,\ldots,n\}$. Thus, $c < \frac{\Delta_M^{CS}(b)}{12}$ implies $\Delta_M^{CS}(b) > \Delta_M$. Also, by SR'_q, $q^0(c,\bar{q},\Delta_i) = 0$ for all $i \in I$.

Accordingly, by Proposition 2, $\Delta_{i+1} = \Delta_i + 4b$ for all $i \in I \setminus \{n\}$, implying $\Delta_M = \Delta_n = \frac{1}{n} + 2(n-1)b$, $1 = \sum_{i\in I}\Delta_i = n\Delta_1 + 2n(n-1)b$.

Then, if $n > N^{CS}(b)$, since $1 = n\Delta_1 + 2n(n-1)b$ and $\Delta_1 > 0$, $2n(n-1)b < 1$, contradicting the definition of $N^{CS}(b)$.

Next, if $n = N^{CS}(b)$, then $\Delta_M = \frac{1}{n} + 2(n-1)b = \frac{1}{N^{CS}(b)} + 2[N^{CS}(b)-1]b = \Delta_M^{CS}(b)$, contradicting $\Delta_M^{CS}(b) > \Delta_M$.

Claim 2. *For any $k \in \mathbb{N}_+$ so that $k < N^{CS}(b)$, $\frac{1}{k} + 2(k-1)b > \frac{1}{k+1} + 2kb$.*

Proof of Claim 2

Suppose not, i.e., there exists $k \in \mathbb{N}_+$ so that $k < N^{CS}(b)$ and $\frac{1}{k} + 2(k-1)b \leq \frac{1}{k+1} + 2kb$. Accordingly, $2k(k+1)b \geq 1$.

Then, by definition of $N^{CS}(b)$, $2k(k+1)b \geq 1 > 2N^{CS}(b)[N^{CS}(b)-1]b$, implying $[N^{CS}(b)+k-1](N^{CS}(b)-k) < 0$, contradicting $k < N^{CS}(b)$. □

By Claim 2, if $n < N^{CS}(b)$, by induction, $\Delta_M = \frac{1}{n} + 2(n-1)b > \frac{1}{N^{CS}(b)} + 2[N^{CS}(b) - 1]b = \Delta_M^{CS}(b)$, contradicting $\Delta_M^{CS}(b) > \Delta_M$.

Thus, each $n \in \mathbb{N}_+$ leads to a contradiction, thus completing the proof of Lemma 4. □

Therefore, to conclude, $\bar{W}^\alpha(b, c', \bar{q}) - \bar{W}^\alpha(b, c, \bar{q}) \geq 0$, and the inequality is strict if $c' < \frac{\Delta_M^{CS}(b)}{12}$ and $\alpha < 1$. □

A.11 Proof of Proposition 6

First, by Lemma 2, there exists some $n \in \mathcal{N}(b, c, \bar{q})$ and some $\Delta \in \mathcal{E}_n(b, c, \bar{q})$ so that $W^\alpha(b, c, \bar{q})|_\Delta = \bar{W}^\alpha(b, c, \bar{q})$.

Also, by Lemma 3, since $0 \leq \bar{q} < \bar{q}' \leq \bar{\bar{q}}$ and $c \geq \frac{b}{3}$, $\Delta \in \mathcal{E}_n(b, c, \bar{q}')$.

Therefore, by part 2 of Corollary 2, $\bar{W}^\alpha(b, c, \bar{q}') - \bar{W}^\alpha(b, c, \bar{q}) \geq 0$. □

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